ASSIGNMENT 1

PROBLEM STATEMENT: -

Set up and Configuration of Hadoop Using CloudEra/ Google Cloud BigQuery. Databricks Lakehouse Platform. Snowflake, Amazon Redshift

OBJECTIVE:

1. Set up a multi-node Hadoop cluster using Cloudera Manager And Configure Hadoop Distributed File System (HDFS) for distributed storage.
2. Create a Google Cloud project and enable BigQuery API and Configure billing and access controls.
3. Create a Databricks workspace and configure clusters.
4. Create a Snowflake account and configure access controls.
5. Launch and configure an Amazon Redshift cluster.

Software Requirements:

Cloudera Manager Server, Google Cloud Platform account and SDK for command-line tools , Databricks account and Apache Spark environment , Snowflake account and UI access , Amazon Redshift cluster , AWS account with permissions and JDBC/ODBC driver.

Hardware Requirements:

• Network connectivity, Storage for Hadoop Distributed File System (HDFS).

• Amazon Redshift, Snowflake is a cloud-based service, Databricks are provides a managed environment, no specific hardware requirements.

THEORY:

Setting up and configuring Hadoop using Cloudera, Google Cloud BigQuery, Databricks Lakehouse Platform, Snowflake, and Amazon Redshift involves different procedures for each platform. Below, I'll provide a brief overview and the general steps for setting up each one. Please note that detailed setup and configuration steps can vary based on the specific version and updates of these platforms.

• **Hadoop using Cloudera:**

1. **Cloudera Distribution**: Download and install Cloudera Distribution for Hadoop (CDH) from the Cloudera website.

2. **Cloudera Manager**: Use Cloudera Manager to set up and manage your Hadoop cluster. Follow the installation and configuration wizard in Cloudera Manager to define hosts, services, and configuration parameters.

3. **HDFS and MapReduce**: Configure Hadoop Distributed File System (HDFS) and MapReduce based on your requirements.
4. **Cluster Deployment**: Deploy the configured cluster.

• **Google Cloud Big Query:**

1.Google Cloud Console: Log in to the [Google Cloud Console](https://console.cloud.google.com/).

2. Enable Big Query API: In the GCP Console, navigate to "APIs & Services" > "Library." Enable the Big Query API.

3. Create a Dataset: Create a BigQuery dataset to organize your tables.

4. Create Tables: Create tables within your dataset, defining the schema and specifying data sources.

5. Access Control: Configure access control for datasets and tables.

• **Databricks Lakehouse Platform:**

1. Databricks Account:

Sign up for a Databricks account: [Databricks](https://databricks.com/trydatabricks).

2. Create a Workspace: Create a Databricks workspace. Configure clusters, libraries, and storage settings.

3. Connect to Data Sources: Connect Databricks to your data sources, such as storage accounts or databases

• **Snowflake:**

1. Snowflake Account:

Sign up for a Snowflake account: [Snowflake](https://www.snowflake.com/free-trial/).

2. Snowflake UI: Use the Snowflake web UI to configure your account, create warehouses, databases, and schemas.

3. Connectivity: Configure network policies and connectivity settings.

4. User and Role Setup: Set up users and roles with appropriate privileges.

• **Amazon Redshift:**

1. **AWS Account**: Sign up for an AWS account: [AWS Signup](https://aws.amazon.com/).
2. **Amazon Redshift Cluster**: In the AWS Management Console, navigate to Amazon Redshift.

Create a new Redshift cluster.

3. **Configure Redshift**: Set up databases, tables, and users as needed. Configure security groups and network settings.

These are general steps, and each platform has detailed documentation that you should follow for the specific versions you are using. Ensure that you follow best practices for security, scalability, and performance in each case.

Applications –

• **Hadoop Using Cloudera :**

1. Data Warehousing.
2. Batch Processing.
3. Machine Learning.

• **Google Cloud BigQuery :**

1. Ad Hoc Analytics
2. Real-time Analytics

• **Databricks Lakehouse Platform:**

1. Unified Analytics.
2. ETL and Data Integration.
3. Data Exploration and Visualization.

• **Snowflake:**

1. Cloud Data Warehousing
2. Data Sharing 3. Time Travel Queries.

• **Amazon Redshift:**

1. Business Intelligence (BI).
2. Data Integration.

Setup - Setting up and configuring Hadoop, Google Cloud BigQuery, Databricks Lakehouse Platform, Snowflake, and Amazon Redshift involves different procedures for each platform. Below, I'll provide you with a brief overview and the steps for setting up each one. Keep in mind that detailed setup and configuration steps can vary based on the specific version and updates of these platforms.

• **Hadoop Setup:**

Download and Install Hadoop:

Visit the Apache Hadoop website and download the version you want to use: [Hadoop Downloads](http://hadoop.apache.org/releases.html) Follow the installation instructions provided with the Hadoop distribution.

Configure Hadoop:

Modify the `core-site.xml`, `hdfs-site.xml`, and other configuration files as needed. Configure the Hadoop environment variables.

Start Hadoop Services:

Use the `start-dfs.sh` and `start-yarn.sh` scripts to start the Hadoop Distributed File System (HDFS) and Yet Another Resource Negotiator (YARN).

• **Google Cloud BigQuery Setup:**

1. **Create a Google Cloud Platform (GCP) Project:**

   Go to the [Google Cloud Console](https://console.cloud.google.com/).
   Create a new project.

2. **Enable BigQuery API:**

   In the GCP Console, navigate to "APIs & Services" > "Library". Enable the BigQuery API.

3. **Set Up Authentication:**

   Create service account credentials with BigQuery access. Download the JSON key file and set the `GOOGLE_APPLICATION_CREDENTIALS` environment variable.

• **Databricks Lakehouse Platform Setup:**

1. Create a Databricks Account:

Sign up for a Databricks account: [Databricks](https://databricks.com/try-databricks)

2. Configure Databricks Workspace:

Create a Databricks workspace. Configure clusters, libraries, and other settings as needed.

3. Connect to Data Sources: Connect Databricks to your data sources, such as storage accounts or databases.

• **Snowflake Setup:**

1.Create a Snowflake Account: Sign up for a Snowflake account: [Snowflake](https://www.snowflake.com/free-trial/)

2. Configure Snowflake: Create a Snowflake warehouse, database, and schema Set up users and roles.

3. Connect to Snowflake: Use a Snowflake client or integration tool to connect to Snowflake.

• **Amazon Redshift Setup:**

1.Create an AWS Account: Sign up for an AWS account: [AWS Signup] (https://aws.amazon.com/)

2.Create an Amazon Redshift Cluster: In the AWS Management Console, navigate to Amazon Redshift. Create a new Redshift cluster.

3.Configure Redshift: Set up databases, tables, and users as needed. Configure security groups and network settings.

Each platform has its own documentation with detailed setup and configuration instructions. Follow the official documentation for the specific versions you are using. Additionally, consider security best practices and compliance requirements when setting up these platforms.

CONCLUSION:

In this way we have Set up and Configuration Hadoop Using CloudEra/ Google Cloud BigQuery, Databricks Lakehouse Platform, Snowflake, Amazon Redshift.

ORAL QUESTION:

1. How did you configure and set up Hadoop within the CloudEra/Google Cloud BigQuery environment?
2. What are the key advantages of using CloudEra or Google Cloud BigQuery for hosting Hadoop?
3. How did you ensure data integrity and security within this environment?
4. What are the key features of Databricks Lakehouse Platform that enhance data processing and analytics capabilities?

ASSIGNMENT 2

PROBLEM STATEMENT: -

Develop a MapReduce program to calculate the frequency of a given word in a given file.

OBJECTIVE:

1. Develop a MapReduce program to calculate the frequency of a given word.
2. Understand the MapReduce programming model and its key components (Mapper and Reducer).
3. Implement the logic for word frequency calculation in the Mapper and Reducer functions.

Hardware Requirements –

- A Hadoop cluster for distributed computing.
- Sufficient computational resources on the cluster to handle the input file.

Software Requirements –

- Hadoop Installation: Ensure that Hadoop is installed and configured on the cluster.
- Java Development Kit (JDK): MapReduce programs are typically written in Java, so JDK is required for development.
- Text Editor or Integrated Development Environment (IDE): Use a text editor or IDE to write and edit the MapReduce program.
- Input File: The file for which the word frequency is to be calculated.

THEORY:

- **Hadoop** is a software framework for easily writing applications which process vast amounts of data (multi-terabyte data-sets) in-parallel on large clusters (thousands of nodes) of commodity hardware in a reliable, fault-tolerant manner. It was popularized by Google and is widely used in open-source implementations like Apache Hadoop.

- **A MapReduce** job usually splits the input data-set into independent chunks which are processed by the map tasks in a completely parallel manner. The framework sorts the outputs of the maps, which are then input to the reduce tasks. Typically both the input and the output of the job are stored in a file- system. The framework takes care of scheduling tasks, monitoring them and re-executes the failed tasks.

• In the **Map phase**, the input data is divided into smaller chunks and processed independently in parallel across multiple nodes in a distributed computing environment. Each chunk is transformed or "mapped" into key-value pairs by applying a user-defined function. The output of the Map phase is a set of intermediate key-value pairs.

• The **Reduce phase** follows the Map phase. It gathers the intermediate key-value pairs generated by the Map tasks, performs data shuffling to group together pairs with the same key, and then applies a user-defined reduction function to aggregate and process the data. The output of the Reduce phase is the final result of the computation.

• The MapReduce framework operates exclusively on pairs, that is, the framework views the input to the job as a set of pairs and produces a set of pairs as the output of the job, conceivably of different types.

• Input and Output types of a MapReduce job: (input) -> map -> -> combine -> -> reduce -> (output)

• **MapReduce** allows for efficient processing of large-scale datasets by leveraging parallelism and distributing the workload across a cluster of machines. It simplifies the development of distributed data processing applications by abstracting away the complexities of parallelization, data distribution, and fault tolerance, making it an essential tool for big data processing in the Hadoop ecosystem.

• **Word Count** is a simple application that counts the number of occurrences of each word in a given input set. This works with a local-standalone, pseudo-distributed or fully-distributed Hadoop installation.

• The text from the input text file is tokenized into words to form a key value pair with all the words present in the input text file. The key is the word from the input file and value is '1'.

• For instance if you consider the sentence "An elephant is an animal". The mapper phase in the Word Count example will split the string into individual tokens i.e. words. In this case, the entire sentence will be split into 5 tokens (one for each word) with a value 1 as shown below – • Key-Value pairs from Hadoop Map Phase Execution-

(an,1)

(elephant,1)

(is,1)

(an,1) (animal,1)

- Hadoop WordCount Example- Shuffle Phase Execution: After the map phase execution is completed successfully, shuffle phase is executed automatically wherein the key-value pairs generated in the map phase are taken as input and then sorted in alphabetical order. After the shuffle phase is executed from the WordCount example code, the output will look like this –

(an,1)

(an,1)

(animal,1)

(elephant,1)

(is,1)

•Hadoop WordCount Example- Reducer Phase Execution: In the reduce phase, all the keys are grouped together and the values for similar keys are added up to find the occurrences for a particular word. It is like an aggregation phase for the keys generated by the map phase. The reducer phase takes the output of shuffle phase as input and then reduces the key-value pairs to unique keys with values added up. In our example "An elephant is an animal." is the only word that appears twice in the sentence. After the execution of the reduce phase of MapReduce WordCount example program, appears as a key only once but with a count of 2 as shown below –

(an, 2)

(animal, 1)

(elephant, 1)

(is, 1)

- This is how the MapReduce word count program executes and outputs the number of occurrences of a word in any given input file. An important point to note during the execution of the WordCount example is that the mapper class in the WordCount program will execute completely on the entire input file and not just a single sentence. Suppose if the input file has 15 lines then the mapper class will split the words of all the 15 lines and form initial key value pairs for the entire dataset. The reducer execution will begin only after the mapper phase is executed successfully.

Code –

Mapper Code –

```
// Importing libraries import java.io.IOException;

import org.apache.hadoop.io.IntWritable; import

org.apache.hadoop.io.LongWritable; import

org.apache.hadoop.io.Text; import

org.apache.hadoop.mapred.MapReduceBase; import

org.apache.hadoop.mapred.Mapper; import

org.apache.hadoop.mapred.OutputCollector; import

org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text,
IntWritable> { // Map function public void map(LongWritable key, Text value,
OutputCollector<Text

IntWritable> output, Reporter rep) throws IOException

{

String line = value.toString();

// Splitting the line on spaces for

(String word : line.split(" "))

{

if (word.length() > 0)

    { output.collect(new Text(word), new

IntWritable(1));

} } } }
```

Reducer Code:

```
// Importing libraries import java.io.IOException; import java.util.Iterator; import
org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.MapReduceBase; import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer; import org.apache.hadoop.mapred.Reporter; public class
WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> { //
Reduce function public void reduce(Text key, Iterator value,
OutputCollector output, Reporter rep) throws IOException
{
int count = 0;
// Counting the frequency of each words while
(value.hasNext())
{
IntWritable i = value.next(); count
+= i.get();
}
output.collect(key, new IntWritable(count));
} }
```

Driver Code:

```
// Importing libraries import java.io.IOException; import
org.apache.hadoop.conf.Configured; import
org.apache.hadoop.fs.Path; import
org.apache.hadoop.io.IntWritable; import
```

```
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.FileInputFormat; import
org.apache.hadoop.mapred.FileOutputFormat; import
org.apache.hadoop.mapred.JobClient; import
org.apache.hadoop.mapred.JobConf; import
org.apache.hadoop.util.Tool; import
org.apache.hadoop.util.ToolRunner; public class WCDriver
extends Configured implements Tool
{    public    int    run(String    args[])    throws
IOException
{
if (args.length < 2)
{
System.out.println("Please give valid inputs"); return
-1;
}
JobConf conf = new JobConf(WCDriver.class);
FileInputFormat.setInputPaths(conf,newPath(args[0]));
FileOutputFormat.setOutputPath(conf,newPath(args[1])); conf.setMapperClass(WCMapper.class);
conf.setReducerClass(WCReducer.class); conf.setMapOutputKeyClass(Text.class);
conf.setMapOutputValueClass(IntWritable.class); conf.setOutputKeyClass(Text.class);
conf.setOutputValueClass(IntWritable.class); JobClient.runJob(conf); return 0;
}
```

```
// Main Method public static void main(String args[])

throws Exception

{  int  exitCode  =  ToolRunner.run(new  WCDriver(),

args);

System.out.println(exitCode);

}

}
```

CONCLUSION:

In this way we have develop a MapReduce program to calculate the frequency of a given word in a given file.

ORAL QUESTION:

1. What are the key phases in a MapReduce program?
2. How do you handle intermediate key-value pairs between the Mapper and Reducer?
3. What challenges might arise when dealing with very large files in MapReduce?
4. Can you explain the concept of shuffling and sorting in MapReduce and its relevance here?

ASSIGNMENT 3

PROBLEM STATEMENT:

Develop a MapReduce program to find the grades of students.

OBJECTIVE:

1. Implement a MapReduce program to calculate grades for students based on their scores.
2. Understand the distributed processing nature of MapReduce in the context of grading.

Hardware Requirements

• A cluster of machines for distributed processing (number of nodes depends on the scale of data).

Software Requirements:

• Hadoop (or any other MapReduce framework).

Data set

• Student data with scores (e.g., CSV file with student ID, name, and scores in different subjects).

Libraries or modules used

• Hadoop MapReduce libraries.

Theory

• **MapReduce Paradigm**

MapReduce is a programming model and processing technique designed for large-scale data processing across distributed clusters. It divides a computational task into two main phases: the Map phase and the Reduce phase. In the context of grading students, MapReduce can efficiently handle the computation of average scores and assignment of grades across a large dataset.

• **Mapper Function**

The mapper function is responsible for processing each input record and emitting intermediate keyvalue pairs. In the case of student grading, the input data likely consists of records in the form of "student_name, subject, score." The mapper function extracts the relevant information, specifically the student name, subject, and score, and emits key-value pairs where the student name is the key and a tuple containing the subject and score is the value.

• **Shuffling and Sorting**

The framework automatically shuffles and sorts the intermediate key-value pairs, grouping them by key. In our student grading example, this means that all records related to a specific student are brought together, ready for processing in the next phase.

• **Reducer Function**

The reducer function takes the grouped key-value pairs and performs the necessary computations to determine the final output. In the context of grading students, the reducer calculates the average score for each student by iterating through the tuples of subjects and scores associated with that student. Once the average score is computed, the reducer assigns a grade based on predefined criteria (e.g., A for scores above 90, B for scores between 80 and 90, and so on). The final output of the reducer is a key-value pair where the key is the student name, and the value is the assigned grade.

• **Scalability and Parallel Processing**

One of the key strengths of MapReduce is its ability to scale horizontally, meaning it can efficiently process large datasets by distributing the computation across multiple nodes in a cluster. Each mapper and reducer operates independently on its subset of data, allowing for parallel processing and significantly reducing the time required for computation.

• **Flexibility and Extensibility**

The MapReduce model is flexible and extensible, making it suitable for a wide range of data processing tasks. In the student grading example, the same MapReduce framework can be adapted for different datasets and grading criteria by modifying the mapper and reducer functions.

Algorithm of work

• **Map phase**

Read student data, emit student ID as the key, and scores as values.

• **Reduce phase**

Calculate grades based on scores and emit the final result.

Applications

• Use cases include automated grading systems for educational institutions.

Code

1. Mapper:

Public class Mapper extends Mapper

{

Public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {

String line = value.toString();

String[] fields = line.split(","); // assuming comma-separated values String

studentId = fields[0];

int marks = Integer.parseInt(fields[1]); // assuming second field is marks

Text outputKey = new Text(studentId);

IntWritable outputValue = new IntWritable(marks); context.write(outputKey,

outputValue); } }

This mapper reads each line of the input data containing student ID and marks separated by a comma.

It then:

- Splits the line into fields based on the comma separator.
- Extracts the student ID and parses the marks as an integer.
- Creates a key-value pair where the key is the student ID and the value is the marks.
- Emits the key-value pair to the reducer.

2. Reducer: Java public class

Reducer extends Reducer

{

public void reducer (Text key, Iterable values, Context context) throws IOException, InterruptedException {

int totalMarks = 0; for

(IntWritable mark : values)

{ totalMarks +=

mark.get();

```
}
String grade = calculateGrade(totalMarks); // Replace with your logic for calculating grades
context.write(key, new Text(grade));
} private String calculateGrade(int
totalMarks) { // Replace with your specific
grade calculation logic if (totalMarks >= 90)
{ return
"A"; } else if
(totalMarks >= 80) {
return "B"; } else
if (totalMarks >= 70) { return
"C"; } else
if (totalMarks >= 60) {
return "D"; } else {
return "F";
} } }
```

This reducer receives key-value pairs from the mapper, where the key is the student ID and the values are individual marks. It:

- Iterates through the marks for each student and calculates the total marks.
- Uses your logic (replace calculateGrade(int totalMarks)) to determine the grade based on the total marks.
- Creates a new key-value pair where the key is the student ID and the value is the calculated grade.

3. Driver: public

```
class Driver
{
Public static void main(String[] args) throws Exception {
Configuration conf = new Configuration(); Job job =
Job.getInstance(conf, "Student Grades");
job.setJarByClass(Driver.class);
```

```
job.setMapperClass(Mapper.class);
job.setReducerClass(Reducer.class);
job.setInputFormatClass(TextInputFormat.class); job.setOutputFormatClass(TextOutputFormat.class);
FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job,newPath(args[1])); System.exit(job.waitForCompletion(true) ? 0 :
1);
}
}
```

This driver configures and submits the MapReduce job:

- Sets the job name and JAR file containing the mapper and reducer classes.
- Specifies the mapper and reducer classes to be used.
- Defines the input format (text file) and output format (text file).
- Sets the input and output paths for the job based on command-line arguments.
- Submits the job and waits for completion, exiting with an appropriate code.

4. **Running the program**:

- Save the code in separate files for Mapper, Reducer, and Driver.
- Compile the files using javac command.
- Run the program using the hadoop jar command, specifying the JAR file, input path, and output path as arguments.
- The output file will contain student IDs and their respective grades.

5. Adapting the program:

- Replace the comma-separated assumption with your actual data format.
- Modify the calculateGrade method to reflect your specific grading system.
- Adjust the input and output formats based on your needs.

CONCLUSION:

In this way we have develop a MapReduce program to find the grades of students.

ORAL QUESTION:

1. How do you handle scenarios where the grading criteria need to be adjusted or updated?
2. How do you ensure the correctness and efficiency of the MapReduce program?
3. What would be the output format of this MapReduce job?

ASSIGNMENT 4

PROBLEM STATEMENT

Mongo DB: Installation and Creation of database and Collection CRUD Document: Insert, Query, Update and Delete Document

OBJECTIVE

1. Understand the installation process of MongoDB and its basic configuration settings.
2. Learn how to create databases and collections in MongoDB to organize data efficiently.
3. Master the CRUD operations (Create, Read, Update, Delete) for managing documents within MongoDB collections.

THEORY

MongoDB Overview: Introduction to MongoDB as a NoSQL document-oriented database, highlighting its advantages and use cases.

Installation Process: Explanation of the steps required to download, install, and configure MongoDB on various operating systems.

Database and Collection Creation: Description of how to create databases and collections in MongoDB using the MongoDB Shell or graphical user interfaces (GUIs) like MongoDB Compass.

CRUD Operations: Overview of the four fundamental CRUD operations in MongoDB:

- Insert: Adding new documents to a collection.

- Query: Retrieving documents from a collection based on specified criteria.

- Update: Modifying existing documents in a collection.

- Delete: Removing documents from a collection.

Step 1: MongoDB Installation on Windows: Download the MongoDB Community Server from the MongoDB Download Center. Run the installer and follow the setup wizard. Add MongoDB's bin folder to the PATH environment variable for easy commandline access.

https://www.mongodb.com/try/download/community

Step 2: Create a Database and Collection:

Switch to Your New Database:

- use myNewDatabase Create a Collection by Inserting a Document:

- 48 db.myNewCollection.insertOne({name: "John Doe", age: 30}) MongoDB creates the database and collection upon inserting the first document.

Step 3: CRUD Operations

Create (Insert Document): Insert a single document:

- db.myNewCollection.insertOne({name: "Jane Doe", age: 25})

- Read (Query Document): Find one document: db.myNewCollection.findOne({name: "John Doe"})

- Update Document: Update a single document: db.myNewCollection.update One ({name: "John Doe"}, {$set: {age: 31}})

Delete Document: Delete a single document:

db.myNewCollection.deleteOne({name: "Bob"})

CONCLUSION

The guide provides a step-by-step approach to installing MongoDB, creating databases and collections, and performing CRUD operations on documents. By mastering these fundamental operations, users can harness the power and flexibility of MongoDB for storing and managing data efficiently. This serves as a foundation for further exploration of MongoDB's advanced features and capabilities in application development and data management.

ORAL QUESTION

1. How do you create a new database in MongoDB?
2. What are the common data types supported in MongoDB documents?
3. What does CRUD stand for in the context of databases?

ASSIGNMENT No: 01

Title: Import Data from different Sources such as (Excel, Sql Server, Oracle etc.) and load in targeted system.

Problem Statement: Import Data from different Sources such as (Excel, Sql Server, Oracle etc.) and load in targeted system.

Prerequisite:

Basics of Python

Software Requirements: Power BI Tool

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn to import data from different sources such as(Excel, Sql Server, Oracle etc.) and load in targeted system.

Outcomes:

After completion of this assignment students are able to understand how to import data from different sources such as(Excel, Sql Server, Oracle etc.) and load in targeted system.

Theory:

What is Legacy Data?

Legacy data, according to Business Dictionary, is "information maintained in an old or outof-date format or computer system that is consequently challenging to access or handle."

Sources of Legacy Data

Where does legacy data come from? Virtually everywhere. Figure 1 indicates that there are many sources from which you may obtain legacy data. This includes existing databases, often relational, although non-RDBs such as hierarchical, network, object, XML, object/relational databases, and NoSQL databases. Files, such as XML documents or "flat files"ù such as configuration files and comma-delimited text files, are also common sources of legacy data. Software, including legacy applications that have been wrapped (perhaps via CORBA) and legacy services such as web services or CICS transactions, can also provide access to existing information. The point to be made is that there is often far more to gaining access to legacy data than simply writing an SQL query against an existing relational database.

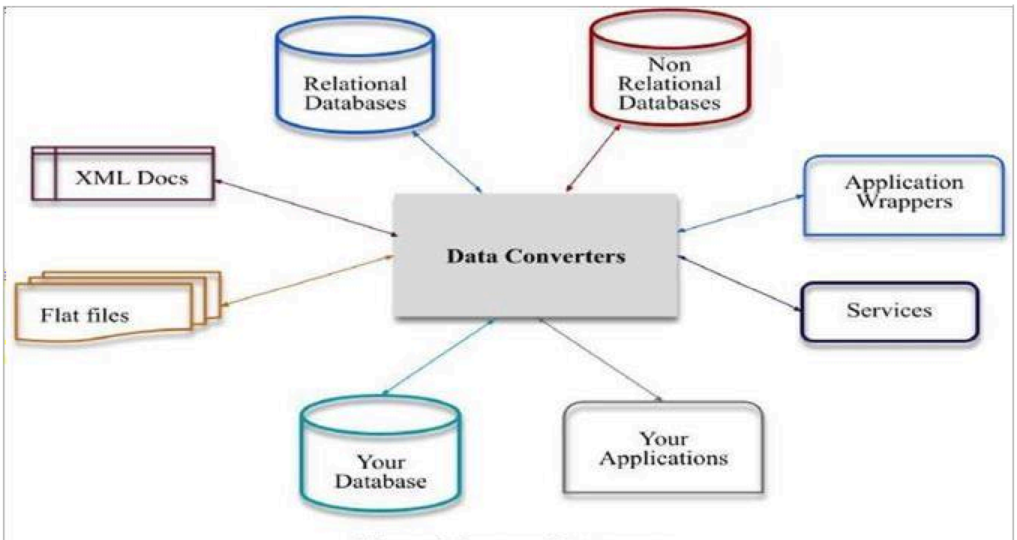
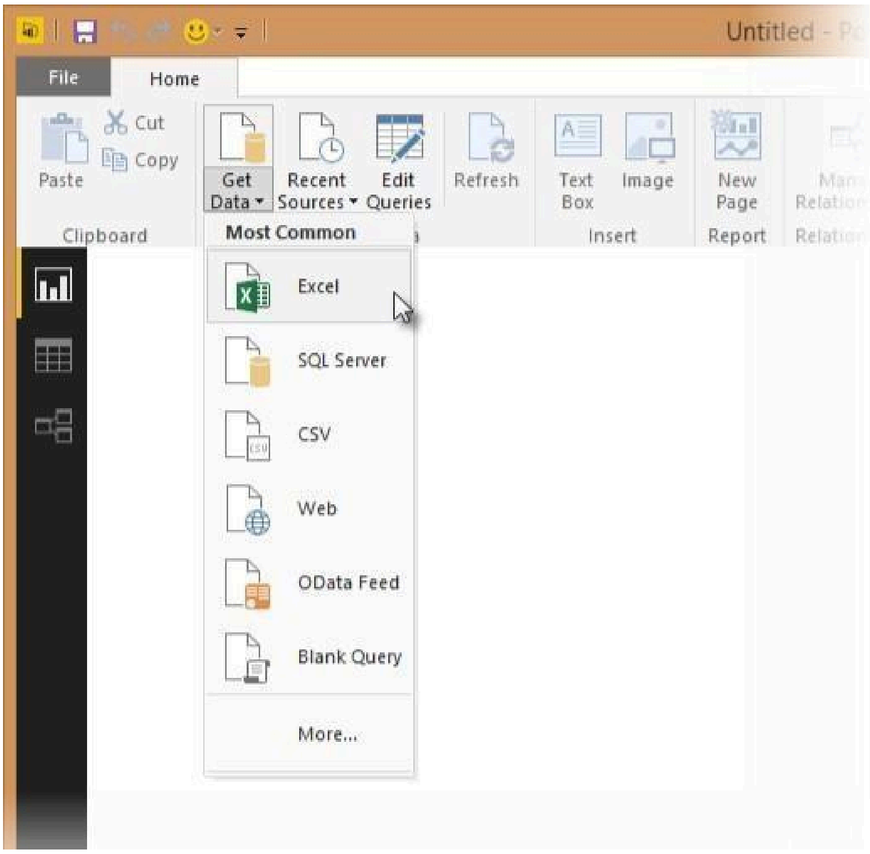


Figure 1. Legacy data sources.

Importing Excel Data

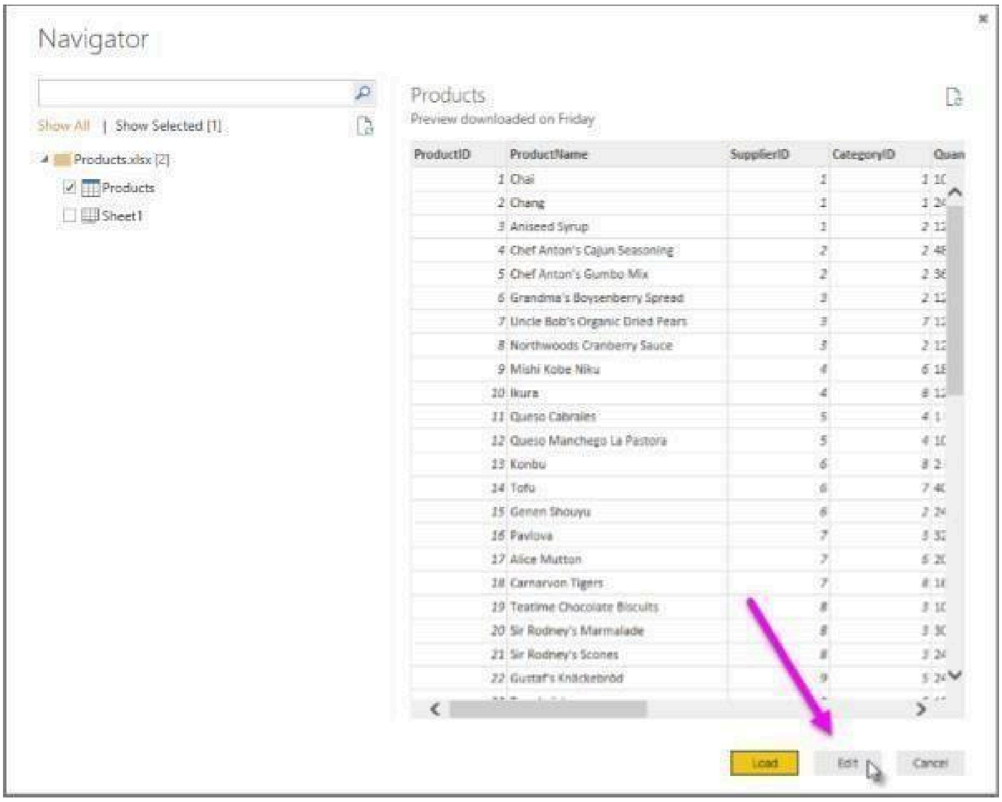
1) Launch Power BI Desktop.
2) From the Home ribbon, select Get Data. Excel is one of the Most Common data connections, so you can select it directly from the Get Data menu.



3) If you select the Get Data button directly, you can also select FIle > Excel and select Connect.

4) In the Open File dialog box, select the Products.xlsx file.

5) In the Navigator pane, select the Products table and then select Edit.



Importing Data from OData Feed

In this task, you'll bring in order data. This step represents connecting to a sales system. You import data into Power BI Desktop from the sample Northwind OData feed at the following URL, which you can copy (and then paste) in the steps below: http://services.odata.org/V3/Northwind/Northwind.svc/

Connect to an OData feed:

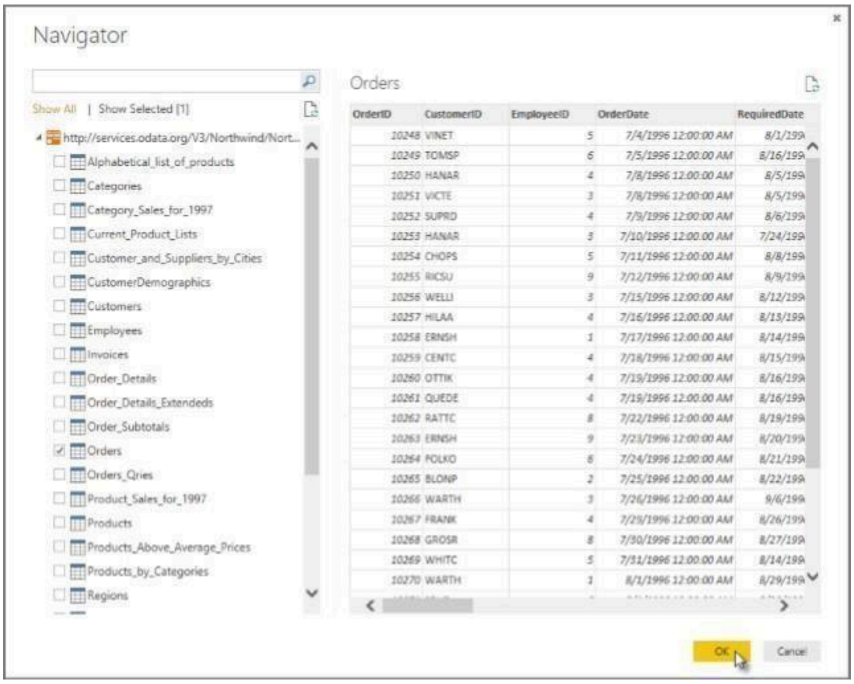
1) From the Home ribbon tab in Query Editor, select Get Data.

2) Browse to the OData Feed data source.

3) In the OData Feed dialog box, paste the URL for the Northwind OData feed.

4) Select OK.

5) In the Navigator pane, select the Orders table, and then select Edit.



Conclusion: - This way, Implemented a program for inverted files.

ASSIGNMENT No: 02

Title: Data Visualization from Extraction Transformation and Loading (ETL) Process.

Problem Statement: Data Visualization from Extraction Transformation and Loading (ETL) Process.

Prerequisite:

Basics of Python

Software Requirements: jupyter

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn Data Visualization from Extraction Transformation and Loading (ETL) Process

Outcomes:

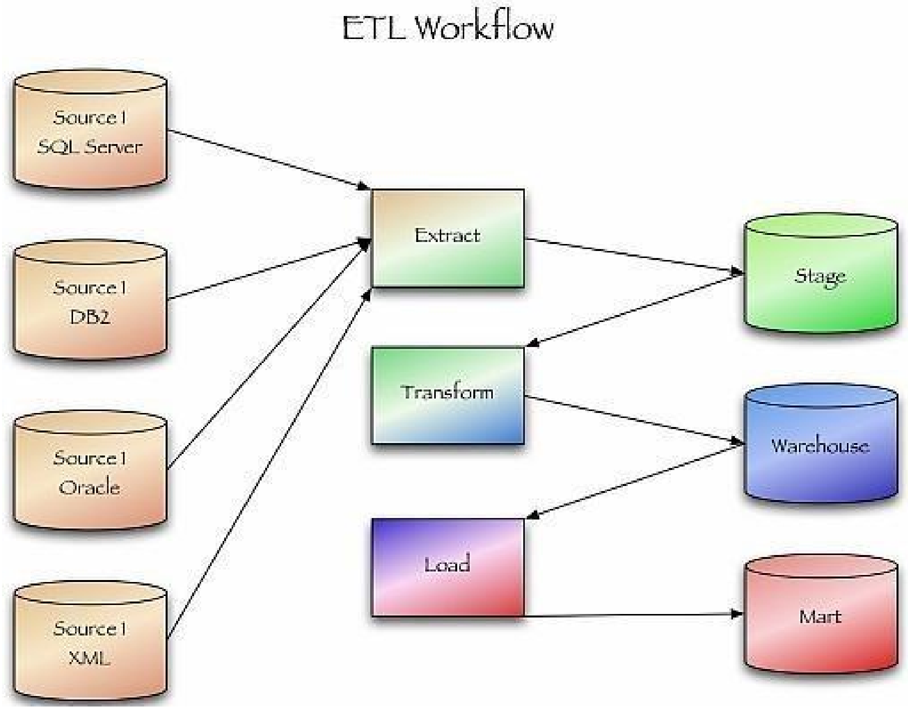
After completion of this assignment students are able to understand how Data Visualization is done through Extraction Transformation and Loading (ETL) Process

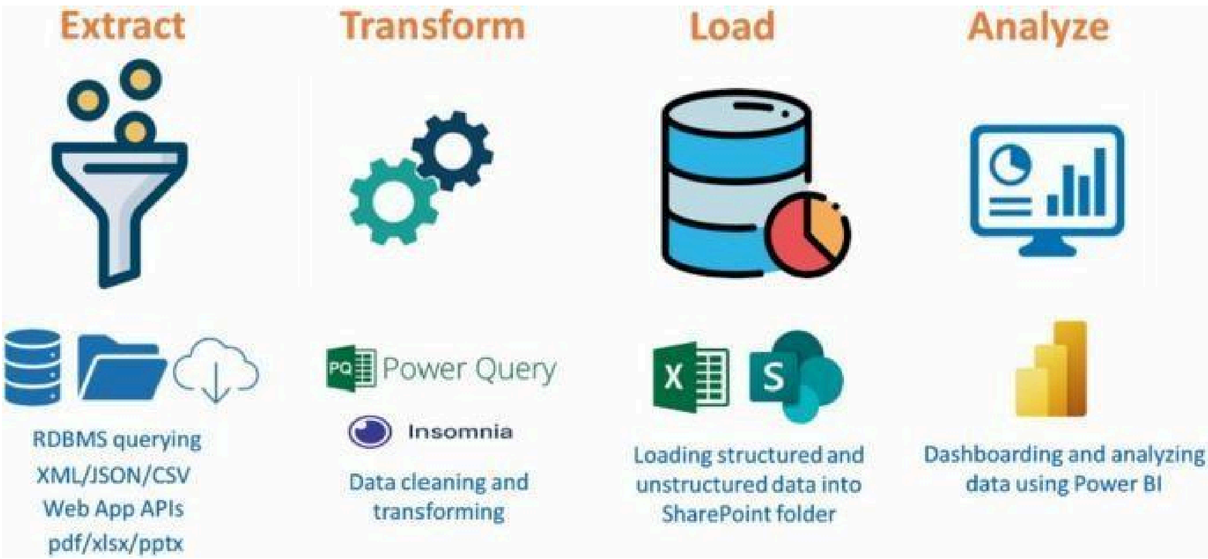
Theory:

Extract, transform, and load (ETL) are 3 data processes, followed after data collection. Extraction takes data, collected in data sources like flat files, databases (relational, hierarchical etc.), transactional datastores, semi-structured repositories (e.g. email systems or document libraries) with different structure and format, pre-validating extracted data and parsing valid data to destination (e.g. staging database)

Transformation takes extracted data and applies predefined rules and functions to it, including selection (e.g. ignore or remove NULLs), data cleansing, encoding (e.g. mapping “Male” to “M”), deriving (e.g. calculating designated value as a product of extracted value and predefined constant) , sorting, joining data from multiple sources (e.g. lookup or merge), aggregation (e.g. summary for each month), transposing (columns to rows or vice versa), splitting, disaggregation, lookups (e.g. validation through dictionaries), predefined validation etc. which may lead to rejection of some data. Transformed data can be stored into Data Warehouse (DW).

Load takes transformed data and places it into end target, in most cases called Data Mart (sometimes they called Data Warehouse too). Load can append, refresh or/and overwrite preexisting data, apply constraints and execute appropriate triggers (to enforce data integrity, uniqueness, mandatory fields, provide log etc.) and may start additional processes, like data backup or replication.





Conclusion:- This way Data Visualization from Extraction Transformation and Loading (ETL) Process is done.

ASSIGNMENT No: 03

Title: Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

Problem Statement: Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

Prerequisite:

Basics of Python

Software Requirements: Jupyter

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn to Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

Outcomes:

After completion of this assignment students are able to understand how to Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

Theory:

Step 1 : Data Extraction :

The data extraction is first step of ETL. There are 2 Types of Data Extraction

1. Full Extraction : All the data from source systems or operational systems gets extracted to staging area. (Initial Load)

2. Partial Extraction : Sometimes we get notification from the source system to update specific date. It is called as Delta load.

Source System Performance: The Extraction strategies should not affect source system performance.

Step 2 : Data Transformation :

The data transformation is second step.After extracting the data there is big need to do the transformation as per the target system.I would like to give you some bullet points of Data Transformation.

- Data Extracted from source system is in to Raw format. We need to transform it before loading in to target server.
- Data has to be cleaned, mapped and transformed.
- There are following important steps of Data Transformation :

1.**Selection :** Select data to load in target

2.**Matching :** Match the data with target system

3.**Data Transforming :** We need to change data as per target table structures

Real life examples of Data Transformation :

- Standardizing data : Data is fetched from multiple sources so it needs to be standardized as per the target system.
- Character set conversion : Need to transform the character sets as per the target systems. (Firstname and last name example)
- Calculated and derived values: In source system there is first val and second val and in target we need the calculation of first val and second val.
- Data Conversion in different formats : If in source system date in in DDMMYY format and in target the date is in DDMONYYYY format then this transformation needs to be done at transformation phase.

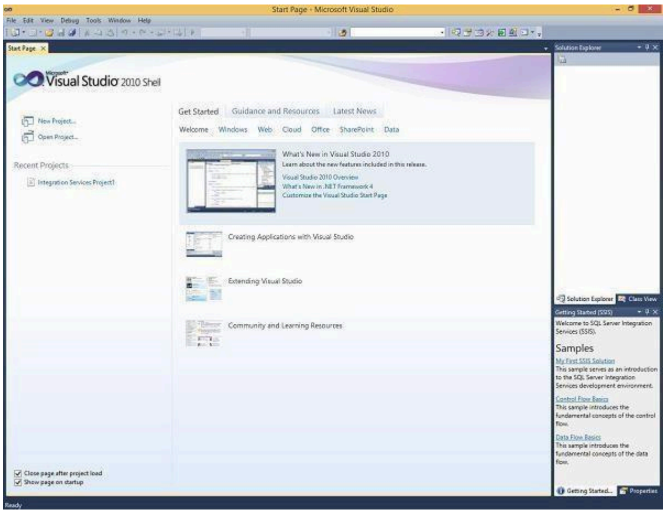
Step 3 : Data Loading

- Data loading phase loads the prepared data from staging tables to main tables.

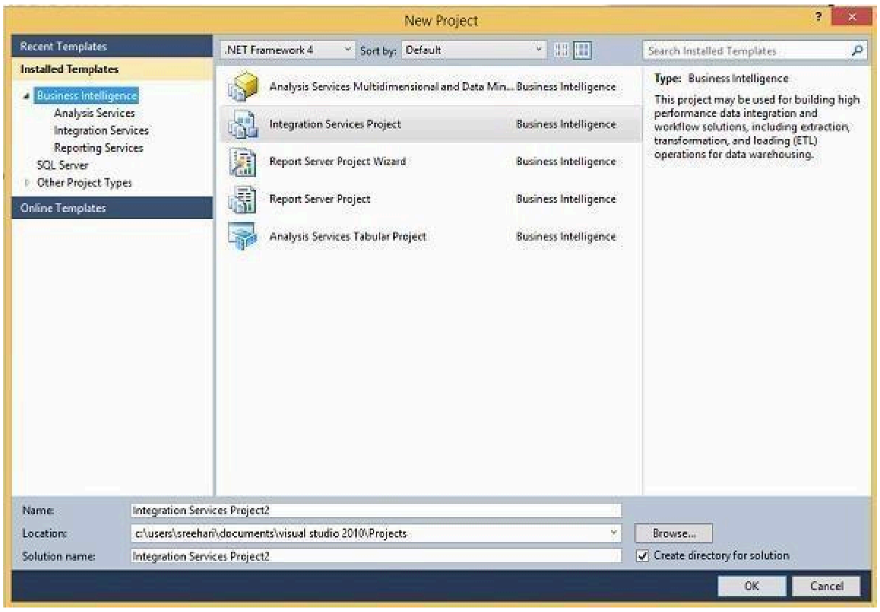
ETL process in SQL Server:

Following are the steps to open BIDS\SSDT.

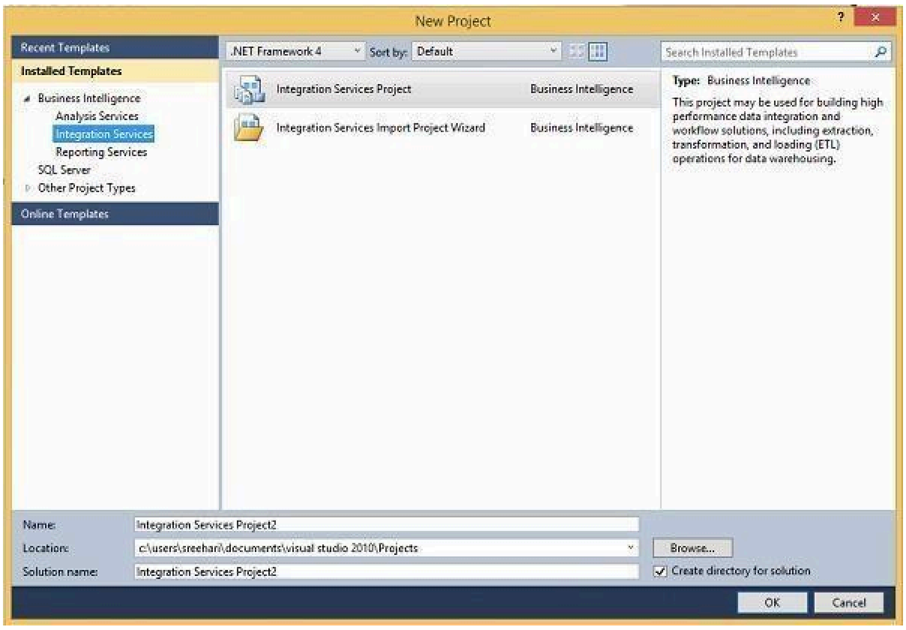
Step 1 − Open either BIDS\SSDT based on the version from the Microsoft SQL Server programs group. The following screen appears.



Step 2 − The above screen shows SSDT has opened. Go to file at the top left corner in the above image and click New. Select project and the following screen opens.



Step 3 − Select Integration Services under Business Intelligence on the top left corner in the above screen to get the following screen.



Step 4 − In the above screen, select either Integration Services Project or Integration Services Import Project Wizard based on your requirement to develop\create the package.

Modes

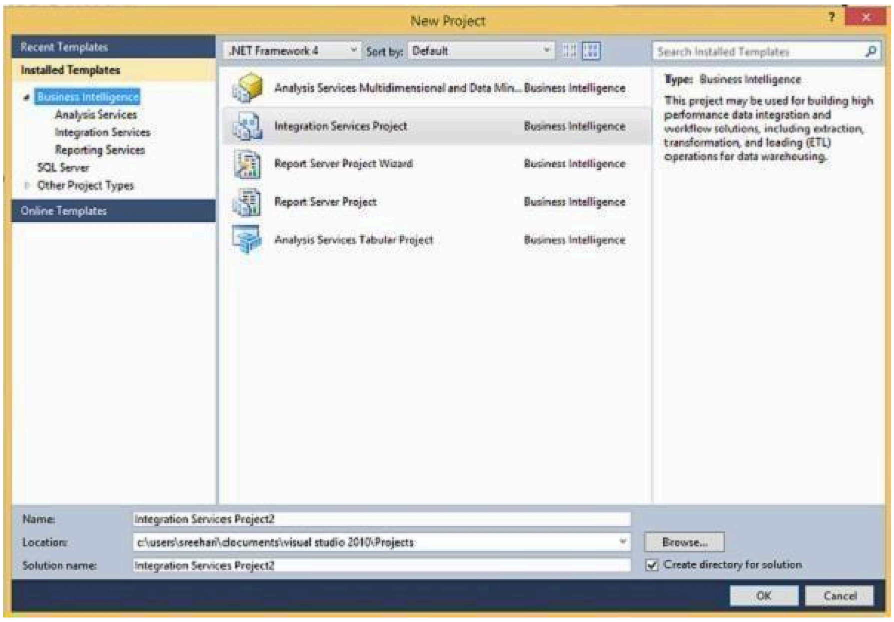
There are two modes − Native Mode (SQL Server Mode) and Share Point Mode.

Models

There are two models − Tabular Model (For Team and Personal Analysis) and Multi Dimensions Model (For Corporate Analysis).

The BIDS (Business Intelligence Studio till 2008 R2) and SSDT (SQL Server Data Tools from 2012) are environments to work with SSAS.

Step 1 − Open either BIDS\SSDT based on the version from the Microsoft SQL Server programs group. The following screen will appear.



Step 4 − In the above screen, select any one option from the listed five options based on your requirement to work with Analysis services.

ETL Process in Power BI

1) Remove other columns to only display columns of interest

In this step you remove all columns except **ProductID**, **ProductName**, **UnitsInStock**, and **QuantityPerUnit**

Power BI Desktop includes Query Editor, which is where you shape and transform your data connections. Query Editor opens automatically when you select **Edit** from Navigator. You can also open the Query Editor by selecting Edit Queries from the Home ribbon in Power BI Desktop. The following steps are performed in Query Editor.

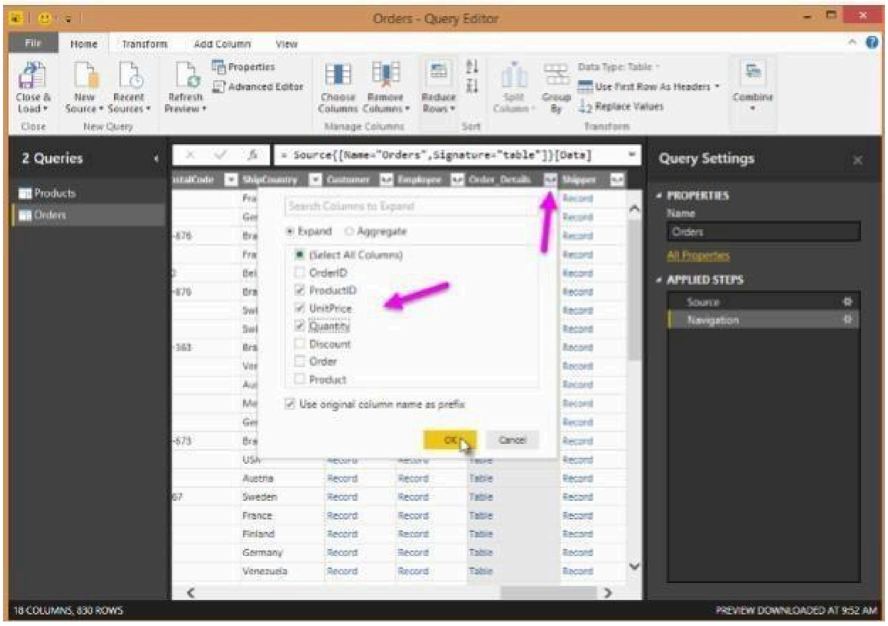
1. In **Query Editor**, select the **ProductID, ProductName**, **QuantityPerUnit,** and **UnitsInStock** columns (use **Ctrl+Click** to select more than one column, or **Shift+Click** to select columns that are beside each other).
2. Select **Remove Columns** > **Remove** Other Columns from the ribbon, or right-click on a column header and click Remove Other Columns.

In this step, you expand the **Order_Details** table that is related to the Orders table, to combine the **ProductID**, **UnitPrice**, and **Quantity** columns from **Order_Details** into the **Orders table**. This is a representation of the data in these tables:

The Expand operation combines columns from a related table into a subject table. When the query runs, rows from the related table (**Order_Details**) are combined into rows from the subject table (**Orders**).

After you expand the Order_Details table, three new columns and additional rows are added to the Orders table, one for each row in the nested or related table.

1. In the Query View, scroll to the Order_Details column.
2. In the Order_Details column, select the expand icon ( ).
3. In the Expand drop-down:
    a. Select (Select All Columns) to clear all columns.
    b. Select ProductID, UnitPrice, and Quantity.
    c. Click OK.



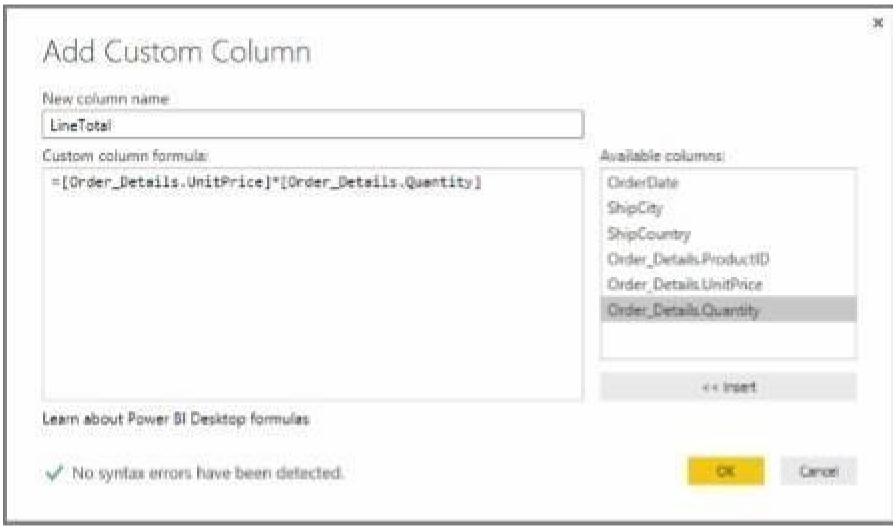
4. Calculate the line total for each Order_Details row

Power BI Desktop lets you to create calculations based on the columns you are importing, so you can enrich the data that you connect to. In this step, you create a Custom Column to calculate the line total for each Order_Details row.

Calculate the line total for each Order_Details row:

1. In the Add Column ribbon tab, click Add Custom Column.

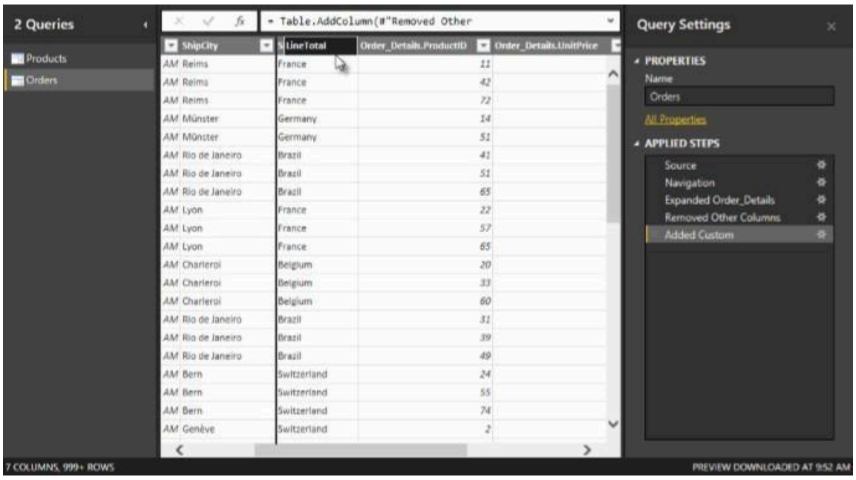
2. In the Add Custom Column dialog box, in the Custom Column Formula textbox, enter [Order_Details.UnitPrice] * [Order_Details.Quantity].
3. In the New column name textbox, enter LineTotal.
4. Click OK.



5. Rename and reorder columns in the query

In this step you finish making the model easy to work with when creating reports, by renaming the final columns and changing their order.

1. In Query Editor, drag the LineTotal column to the left, after ShipCountry.



2.Remove the Order_Details. prefix from the Order_Details.ProductID, Order_Details.UnitPrice and Order_Details.Quantity columns, by double-clicking on each column header, and then deleting that text from the column name.

6. Combine the Products and Total Sales queries

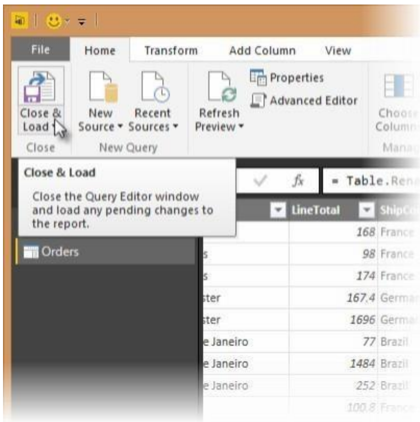
Power BI Desktop does not require you to combine queries to report on them. Instead, you can create Relationships between datasets. These relationships can be created on any column that is common to your datasets

we have Orders and Products data that share a common 'ProductID' field, so we need to ensure there's a relationship between them in the model we're using with Power BI Desktop.Simply specify in Power BI Desktop that the columns from each table are related (i.e. columns that have the same values). Power BI Desktop works out the direction and cardinality of the relationship for you. In some cases, it will even detect the relationships automatically.

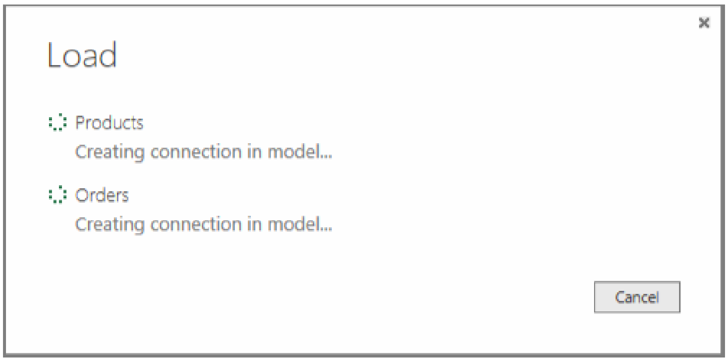
In this task, you confirm that a relationship is established in Power BI Desktop between the Products and Total Sales queries

Step 1: Confirm the relationship between Products and Total Sales

1. First, we need to load the model that we created in Query Editor into Power BI Desktop. From the Home ribbon of Query Editor, select Close & Load.

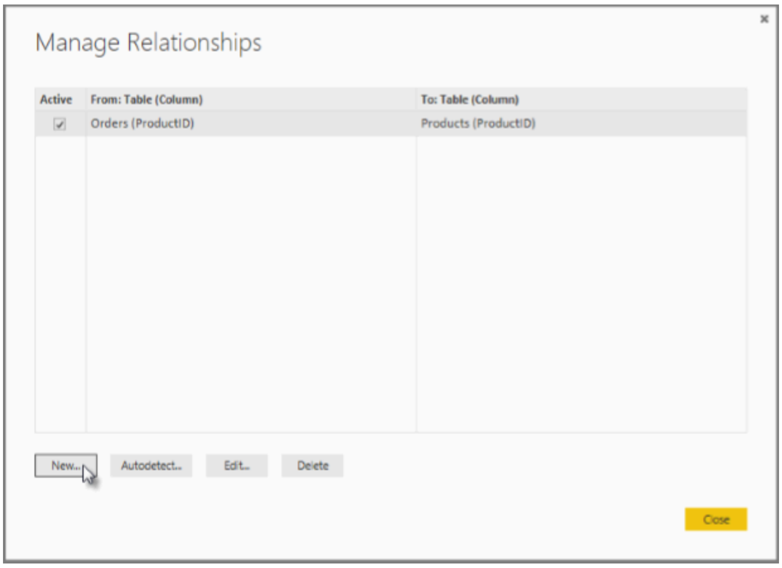


1. Power BI Desktop loads the data from the two queries.

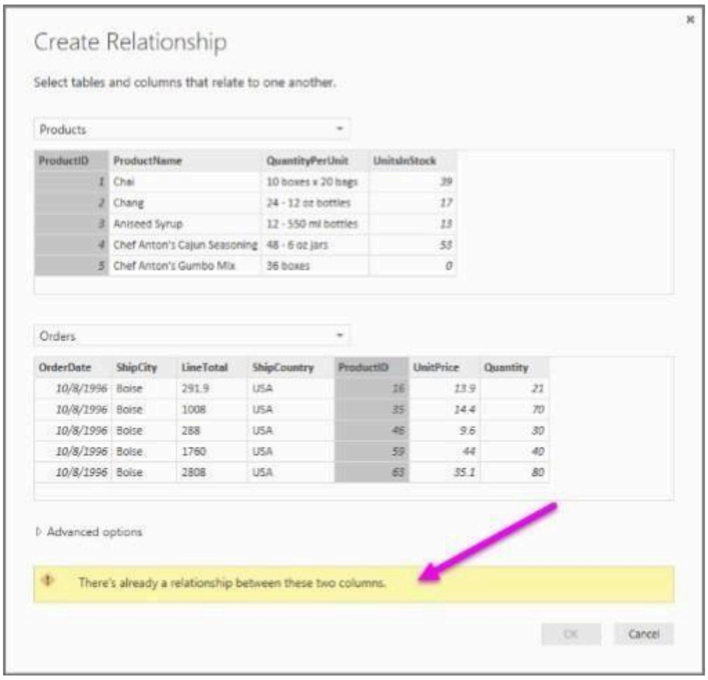


2. Once the data is loaded, select the Manage Relationships button Home ribbon.

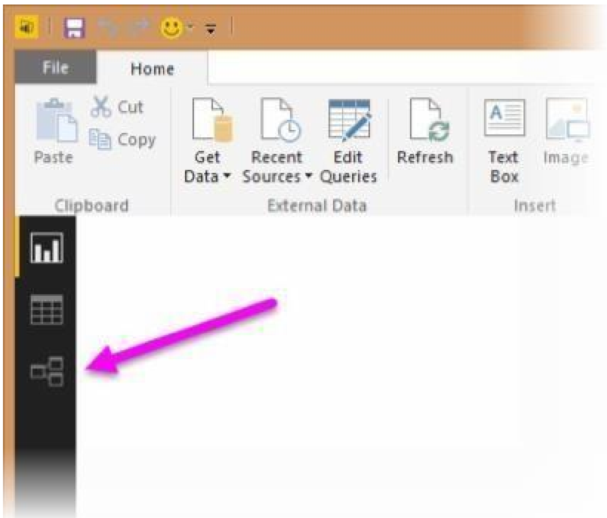
3. Select the New… button



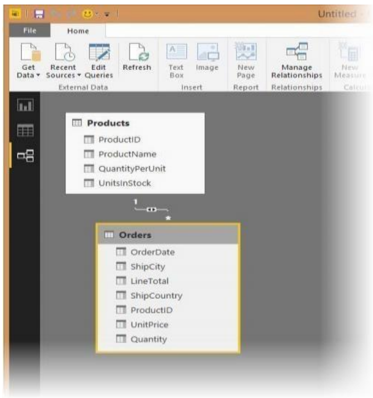
4. When we attempt to create the relationship, we see that one already exists! As shown in theCreate Relationship dialog (by the shaded columns), the ProductsID fields in each query already have an established relationship.



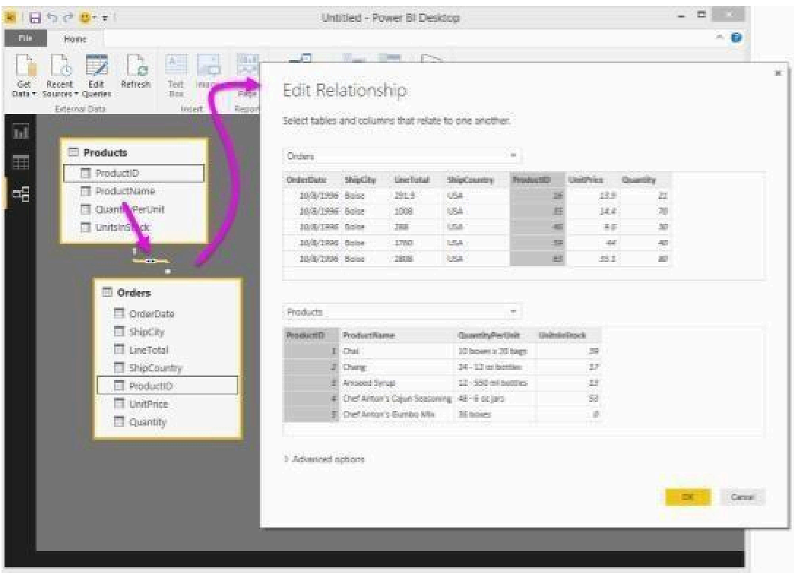
5. Select Cancel, and then select Relationship view in Power BI Desktop.



6. We see the following, which visualizes the relationship between the queries.



7. When you double-click the arrow on the line that connects to the queries, an Edit Relationship dialog appears.



No need to make any changes, so we'll just select Cancel to close the Edit Relationship dialog

Conclusion: Thus Performed Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

ASSIGNMENT No: 4

Title: Data Analysis and Visualization using Advanced Excel.

Problem Statement: Data Analysis and Visualization using Advanced Excel.

Prerequisite:

Basics of Python

Software Requirements: Jupyter

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn to Perform Data Analysis and Visualization using Advanced Excel.

Outcomes:

After completion of this assignment students are able to understand Data Analysis and Visualization using Advanced Excel.

Theory:

Defining Data Visualization

We'll first start by defining what data visualization is. Data visualization is a graphical representation of data. By utilizing charts, graphs, maps, etc., we can provide a simple and accessible way to understand our data and identify trends and outliers within our datasets. Note that Excel uses the term "chart" to mean a "plot". For example, a bar plot is called a bar chart in Excel terminology.

The purpose of this tutorial is to walk you through some basic charts to visualize your data before jumping into more advanced techniques later on. We highly recommend you check out our Data Visualization cheat sheet to learn more about the most common visualizations and when to use them.

Example Dataset

We first need a dataset to work with before creating any visualizations. This tutorial will use a simple dataset containing sales data for a local electronics store. The dataset contains information on the number of units sold for various product types in 2022 and totals for columns and rows.

Month	TVs	Mobile Phones	Laptops	Total
1/1/2022	145	335	82	562
2/1/2022	145	362	126	633
3/1/2022	105	311	95	511
4/1/2022	171	259	93	523
5/1/2022	178	277	107	562
6/1/2022	167	292	145	604
7/1/2022	200	385	77	662
8/1/2022	181	388	78	647

9/1/2022	152	291	83	526
10/1/2022	143	345	102	590
11/1/2022	114	399	99	612
12/1/2022	109	250	101	460
Total	1810	3894	1188	

We'll be working with this dataset throughout this tutorial. You can download the data file from GitHub.

Alternatively, you can import the dataset using the following steps:

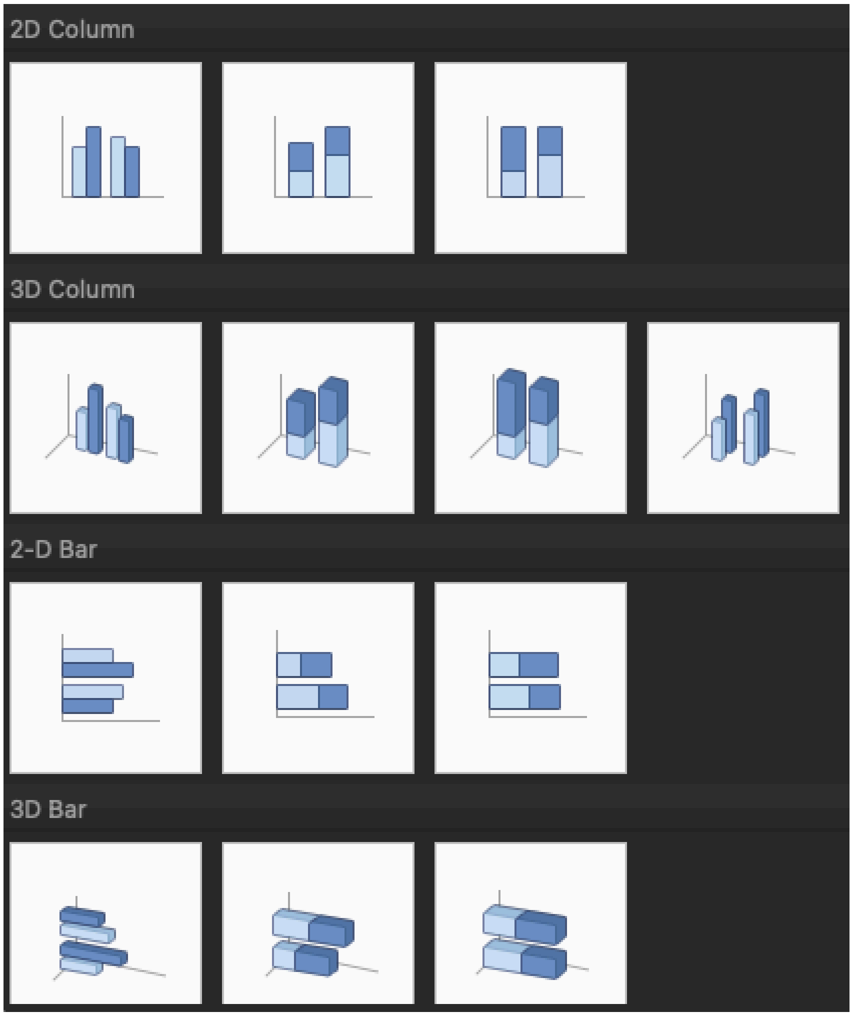
- Open Excel and create a new workbook.
- Copy the dataset above and paste it into cell A1

Format the cells as needed (e.g., adjust column width, apply bold formatting to headers, etc.).

	A	B	C	D
1	**Month**	**Television**	**Laptop**	**Mobile Phones**
2	1/1/2022	145	335	82
3	2/1/2022	145	362	126
4	3/1/2022	105	311	95
5	4/1/2022	171	259	93
6	5/1/2022	178	277	107
7	6/1/2022	167	292	145
8	7/1/2022	200	385	77
9	8/1/2022	181	388	78
10	9/1/2022	152	291	83
11	10/1/2022	143	345	102
12	11/1/2022	114	399	99
13	12/1/2022	109	250	101

Creating Basic Charts in Excel

Excel has multiple options for choosing a particular chart type. For example, if you want to create a column or bar chart, you are often presented with various visualization options. For example, there are 2D and 3D versions and normal, stacked, and 100% stacked options. Depending on your requirements, you can choose the visualization type that best suits your needs.

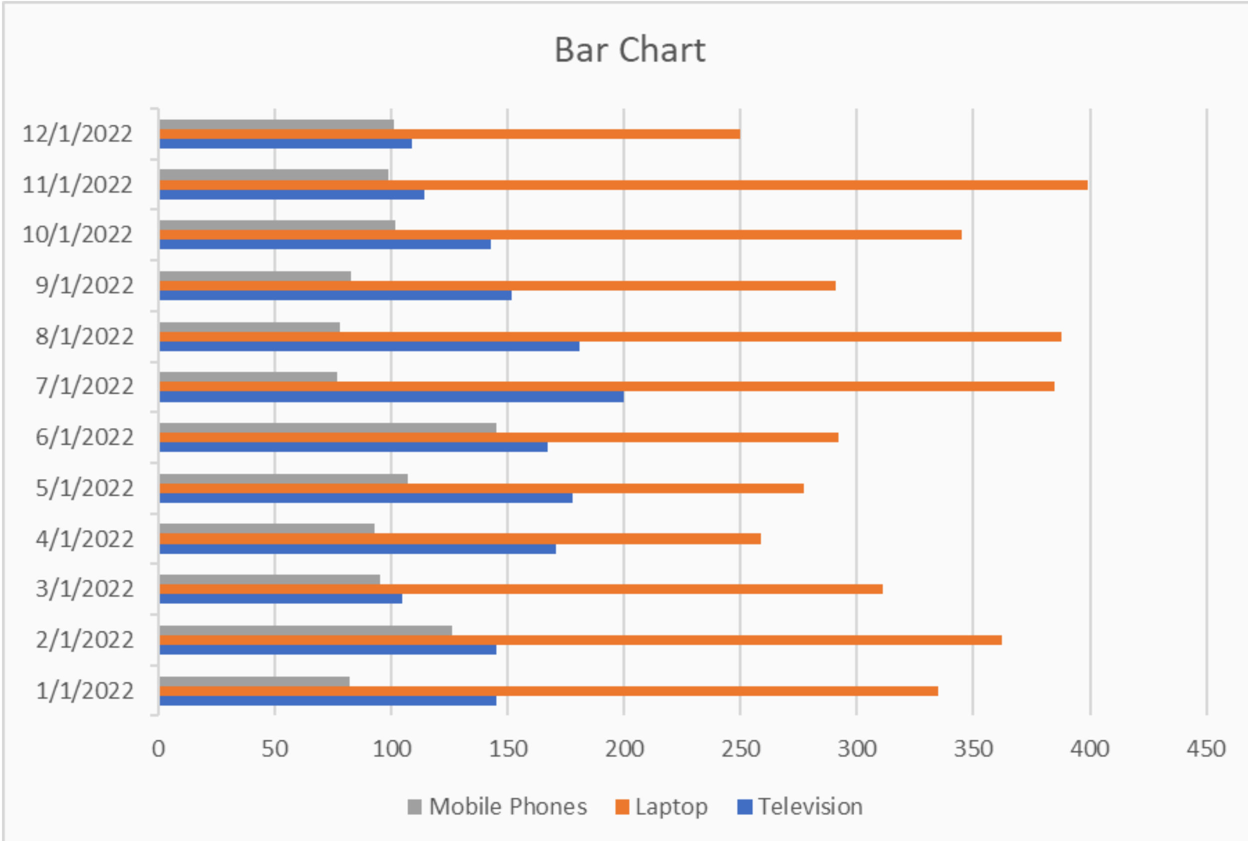


Excel bar charts

Bar charts are one of the easiest charts to interpret, enabling the person viewing the chart an easy way to compare categorical data quickly. On a bar chart, the categorical data is on the y-axis, and the values are on the x-axis.

To create a bar chart:

- Select the data range A1:D13
- Click the "Insert" tab in the Excel ribbon
- Click on the columns icon button dropdown, and under the “2-D Bar” category, choose “Clustered Bar”



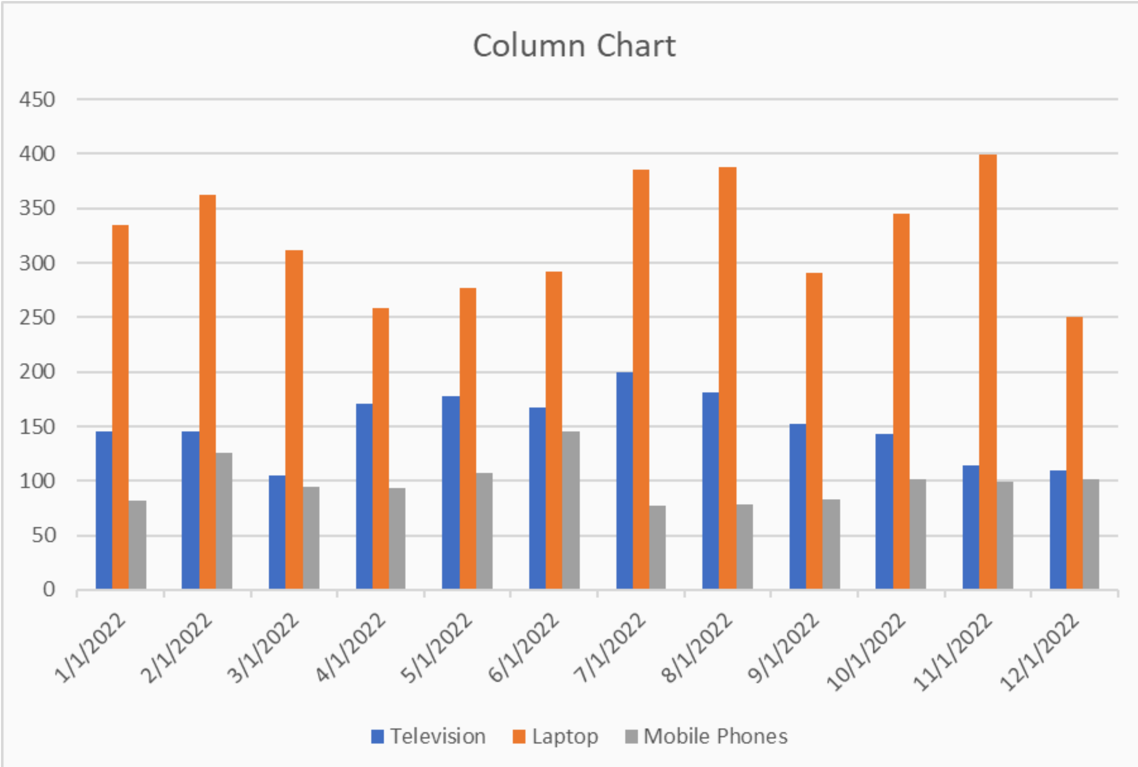
Excel column charts

A column chart, also known as a vertical bar chart, helps visualize data where categories are placed on the x- axis and the values on the y-axis. Similar to bar charts, they help visualize data across categories.

To create a column chart in Excel:

- Select the data range A1:D13
- Click the "Insert" tab in the Excel ribbon
- Click on the columns icon dropdown, and under the “2-D Column” category, choose “Clustered Column”

You can now see a column chart that displays the number of units sold for each product category by the month.



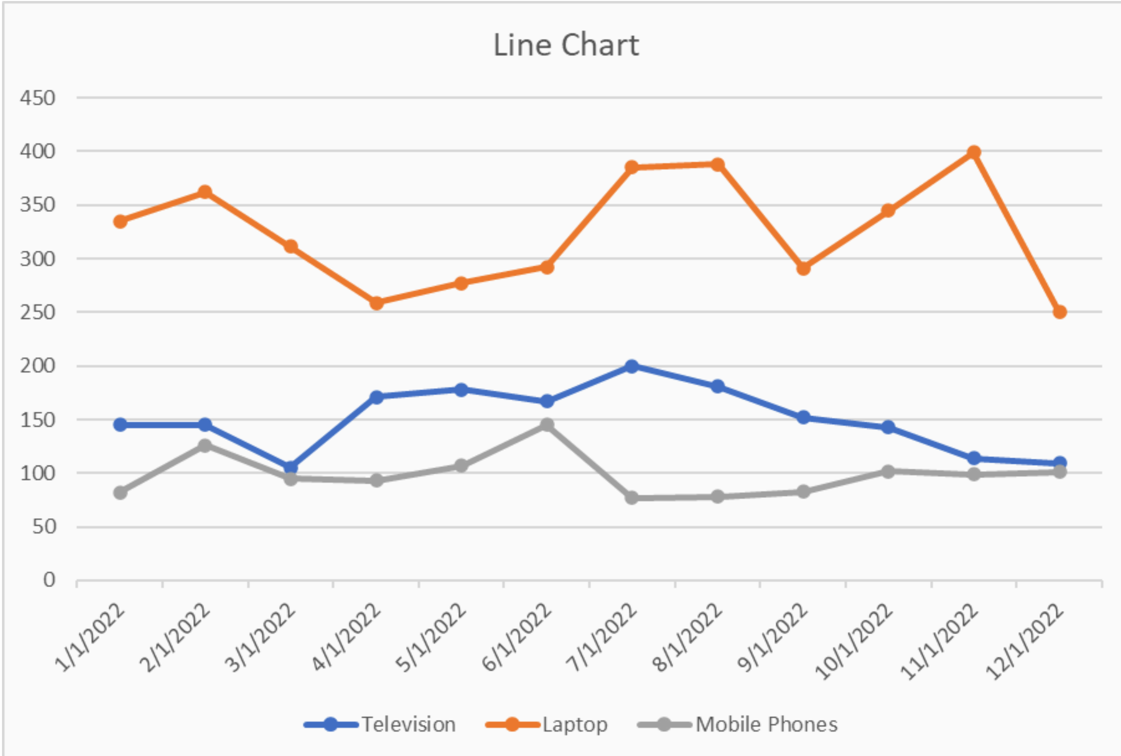
Excel line charts

A line chart is the most useful way to capture how a numerical variable changes over time. This is helpful to identify trends in numeric values.

To create a line chart in Excel:

- Select the data range A1:D13
- Click the "Insert" tab in the Excel ribbon
- Click on the line chart dropdown, and under the “2-D Line” category, choose “Line with Markers"

You can now see a line chart displaying units sold each month split by product category. This enables you to compare each product category's performance over time easily.

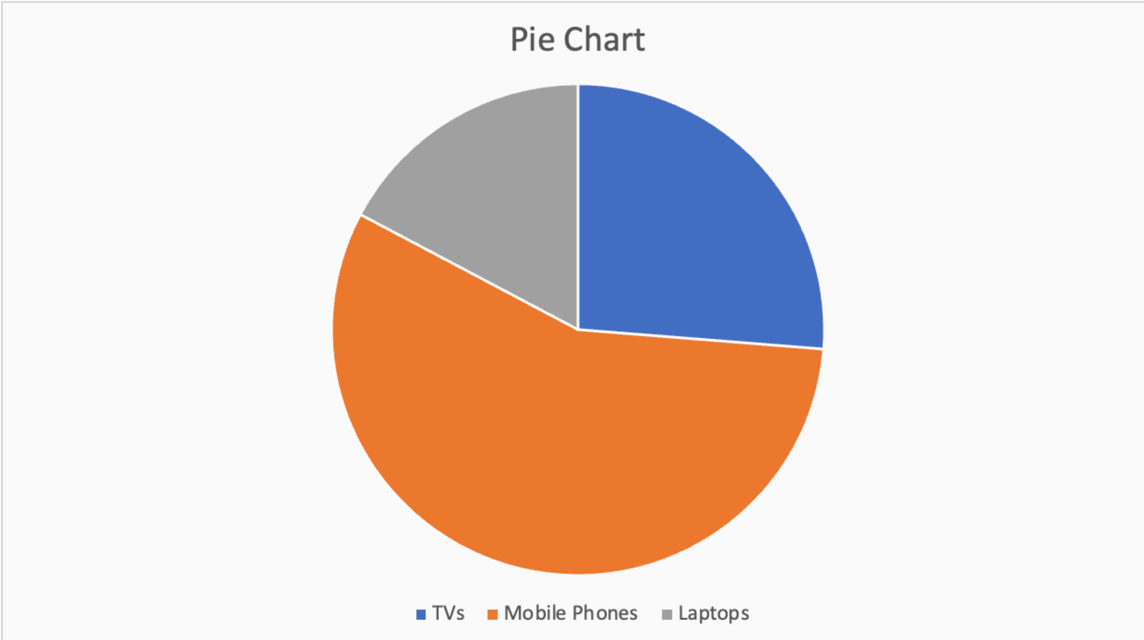


Excel pie charts

A pie chart is most commonly used to show the proportions of a whole. It's like visualizing fractions when you were in high school. With this pie chart, we want to compare the total sales between the three categories.

To create a pie chart in Excel:

- First, select the data range B1:D1
- Second, using the command (for Mac) or ctrl (for Windows), select the second date range: B14:D14
- Click the "Insert" tab in the Excel ribbon
- Click on the pie chart dropdown, and under the "2-D Pie" category, choose "Pie"

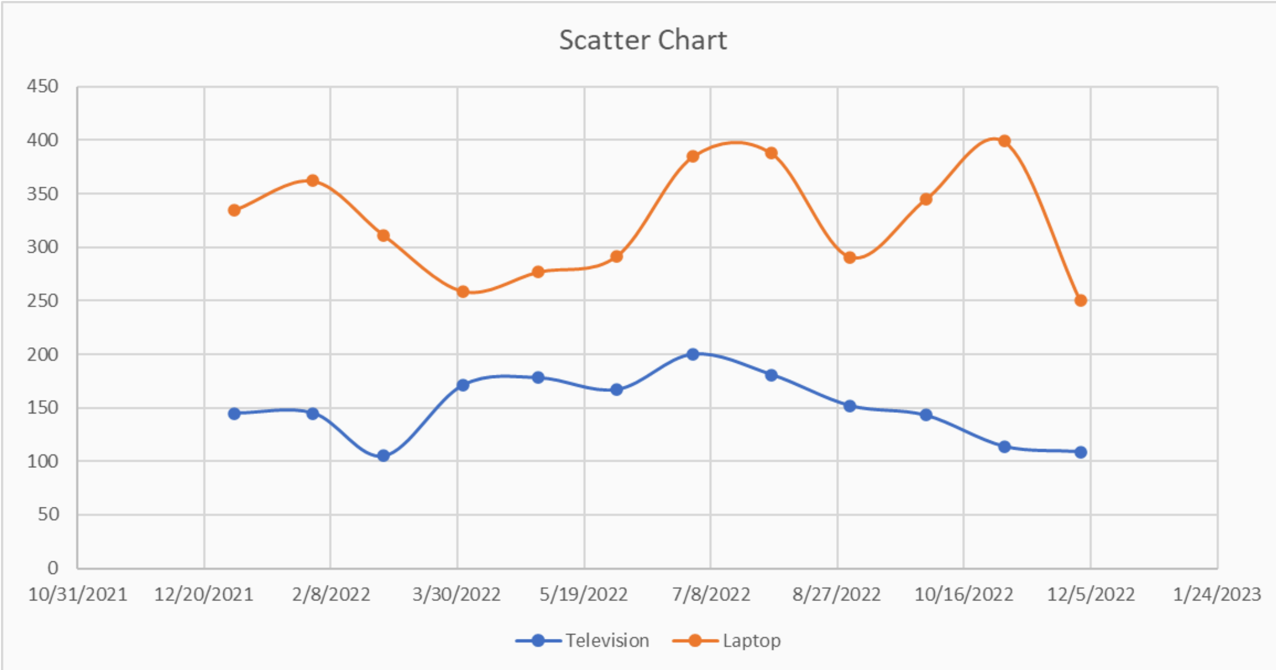


Advanced Excel Visualization Techniques
Excel scatter plots

A scatter plot is commonly used to visualize the relationship between two variables. It can be useful for quickly surfacing potential correlations between data points. We'll create a scatter plot to compare the number of TVs and laptops sold.

To create a scatter plot in Excel:

- Select the data range A1:C13
- Click the "Insert" tab in the Excel ribbon
- Click on the scatter plot dropdown, and under the "Scatter" category, choose "Histogram"
- Click "Scatter or Bubble Chart" and choose "Scatter with Smooth Lines and Markers"



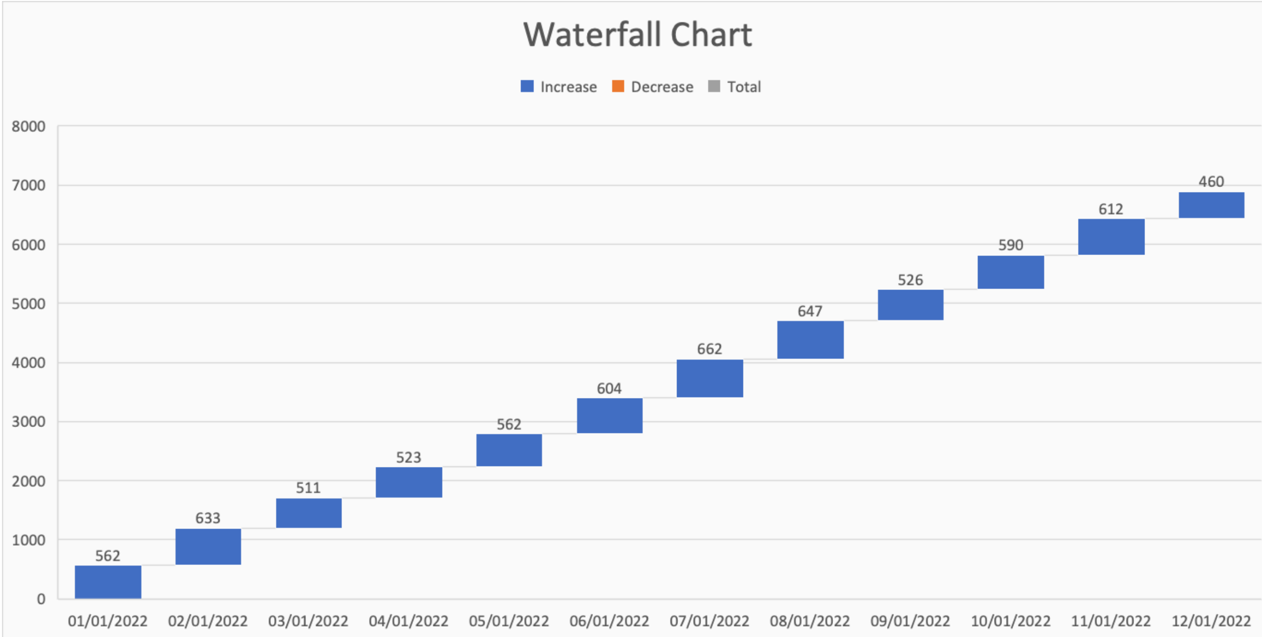
Excel waterfall chart

A waterfall chart is a special chart that helps illustrate how positive and negative values can contribute to a total. They can be great for visualizing changes over time. In our example, we'll compare the total sales for each month regardless of the categories.

To create a waterfall chart in Excel:

- First, select the data range A2:A13
- Second, using the command (for Mac) or ctrl (for Windows), select the second data range E2:E13
- Click on the waterfall chart dropdown, and under the "Waterfall" category, choose "Waterfall"

We'll only see positive values in our example because the Total column only contains positive values, but this chart can be great for comparing financial data and changes over time.



Customizing and Formatting Excel Charts

Aside from creating charts in Excel, there are many options to customize and format a chart. From here, we'll discuss some common formatting options that'll help you enhance your visuals.

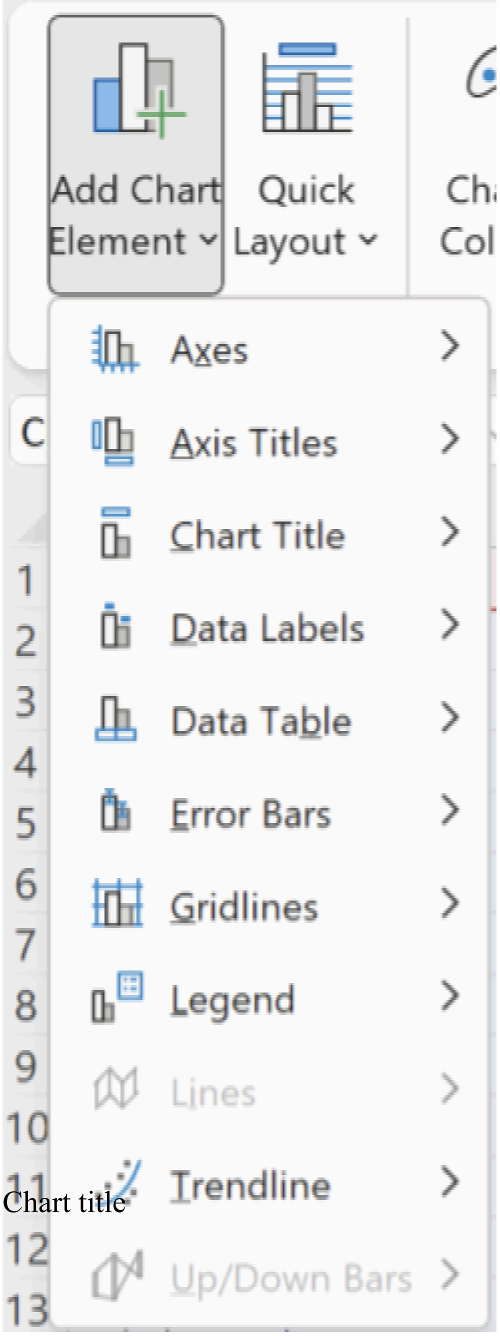
First, create a column chart based on our previous guidance that displays the number of units sold for each product category by the month.

Chart elements

Chart elements include titles, legends, data labels, gridlines, and axes. You can add, remove, or modify these elements as needed.

Click on the chart to select it.

- In the Excel ribbon, click the "Chart Design" tab
- Click the "Add Chart Element" dropdown to access the available chart elements



Add Chart Element
Quick Layout
Axes
Axis Titles
Chart Title
Data Labels
Data Table
Error Bars
Gridlines
Legend
Lines
Trendline
Up/Down Bars

Chart title

- To add a chart title, click "Add Chart Element" > "Chart Title" and choose one of the available options (e.g., "Above Chart" or "Centered Overlay"). You can then click the title placeholder and type your desired title
- To remove a chart title, right-click on the title and select “Delete”

For this tutorial, rename the chart “Electronic Store Sales 2022”

Legend

- To modify the chart legend, click "Add Chart Element" > "Legend" and choose a position for the legend (e.g., "Right" or "Top")
- To remove the legend, click "Add Chart Element" > "Legend" > "None"

For this tutorial, move the legend to the top of the chart.

Data labels

- To add data labels, click "Add Chart Element" > "Data Labels" and choose one of the available options (e.g., "Center" or "Above"). This will display the data values directly on the chart
- To remove data labels, click “Add Chart Element” > “Data Labels” > “None”
- To remove data labels from individual categories, right-click on the data label of the chosen category; this will highlight all relevant data labels. You can then click “Delete”

For this tutorial, add data labels for all categories.

Your chart should now look like this:

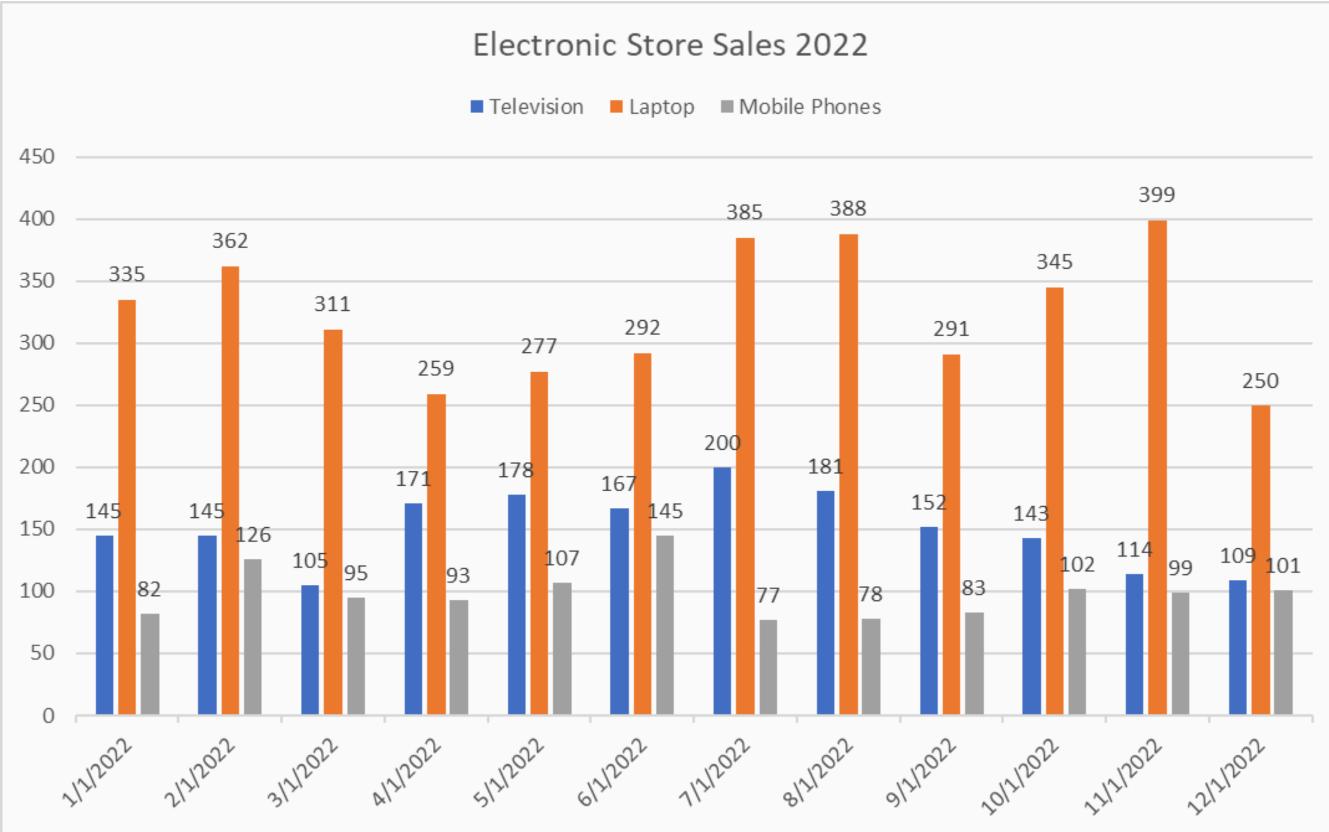
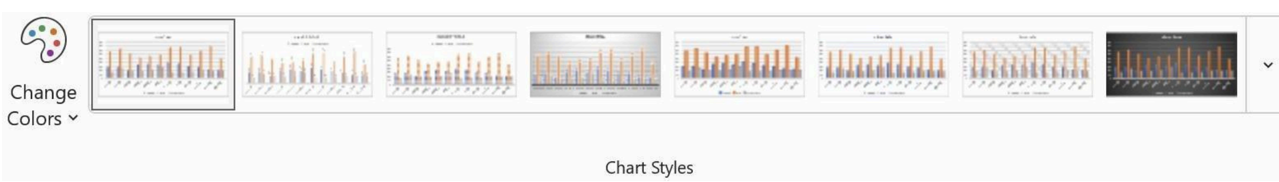


Chart Styles and colors

On top of customizing individual chart elements, you can change your chart's overall look and feel by applying different styles and color schemes.

To customize a chart style, select the visual you want to update. Then take the following steps:

- In the Excel ribbon, click the "Chart Design" tab
- Under the “Chart Styles” section, there is an option to change styles and change colors
- For chart styles: Browse the available styles and click one to apply to your chart
- For color changes: Click the “Change Colors” dropdown and choose a color scheme



By implementing small visual changes, we can see a big difference in the aesthetic of the whole chart. Here's an example of how our chart has changed by choosing Style 6 and Monochromatic Palette 8.

Formatting Excel chart axes

To improve the readability of our chart, we can format our axis in several ways, including axis titles, scale, or even visibility.

Axis titles

- To add an axis title, click "Add Chart Element" > "Axis Titles" and choose "Primary Horizontal" or "Primary Vertical." You can then click the title placeholder and type your desired title
- To remove an axis title, right-click on the axis title and click "Delete"

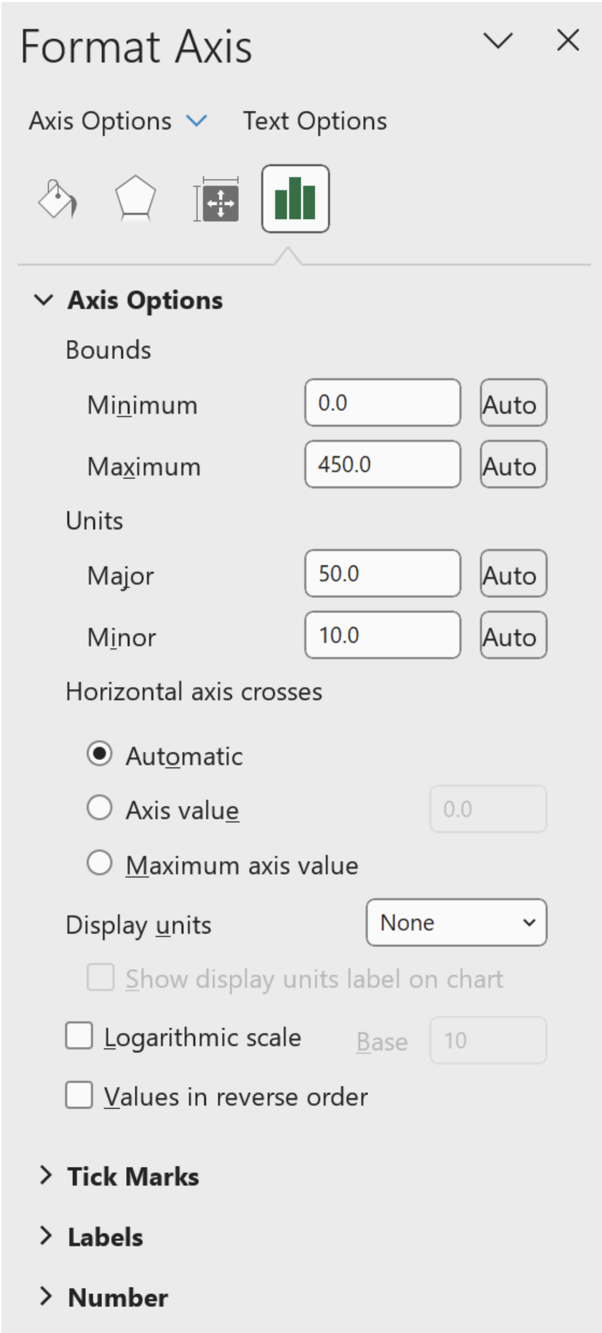
For this tutorial, add an x-axis title "Month" and a y-axis title "Products

Sold." Axis scale and number format

To adjust the axis scale or number format:

- Right-click the axis you want to modify and choose "Format Axis"

In the "Format Axis" pane, you can change the minimum and maximum values, major and minor units, or number format.

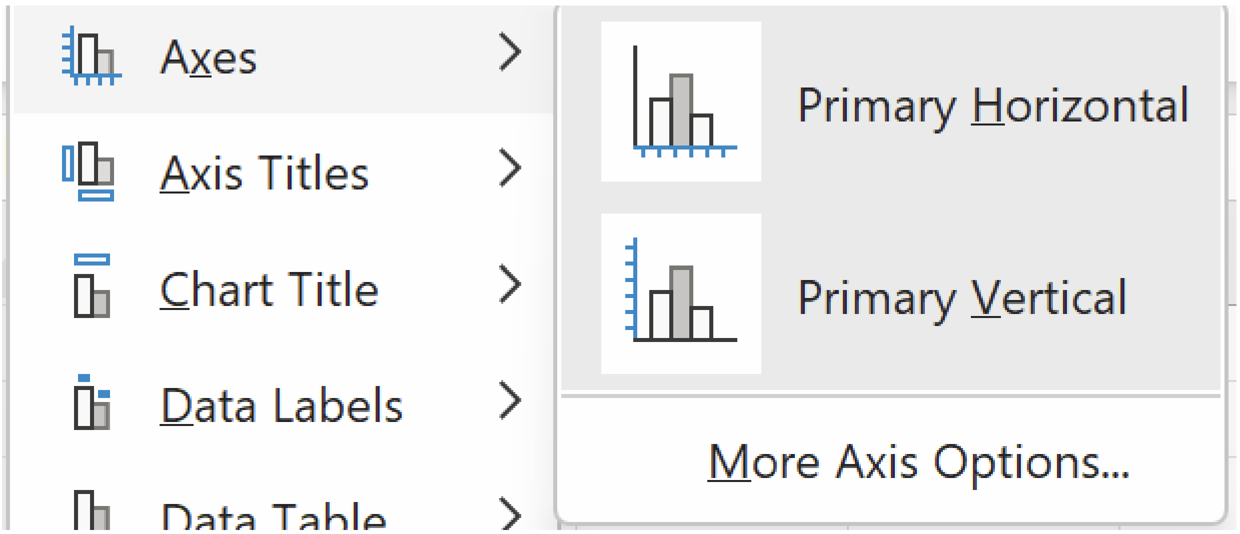


For this tutorial, we intend to keep these options the same but feel free to play around and test what each does.

Axis visibility

Finally, if you would like to remove the axis labels from showing, you can take the following steps:

- In the Excel ribbon, click the "Chart Design" tab
- Click the "Add Chart Element" dropdown and navigate to "Axes"
- By default, both options will have a darker gray box surrounding them, to remove an Axes, simply de-select the axes you'd like to remove

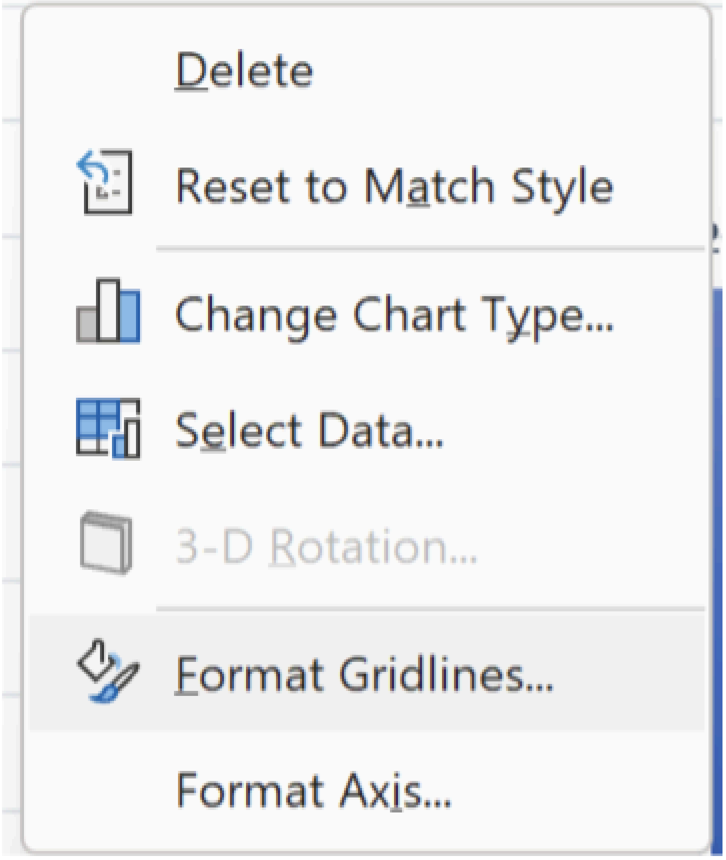


Other Excel formatting options

There are other formatting options available, including modifying data series colors, adjusting chart and plot area backgrounds, and customizing gridlines.

To access these options:

- Right-click the chart element you want to modify (e.g., data series, plot area, or gridlines)
- Select "Format [element]"



Conclusion:- Thus, this way Data Analysis and Visualization is done using Advanced Excel.

ASSIGNMENT No: 5

Title: Perform the data classification algorithm using any Classification algorithm

Problem Statement: Perform the data classification algorithm using any Classification algorithm

Prerequisite:

Basics of Python

Software Requirements: Jupyter

Hardware Requirements:

PIV, 2GB RAM, 500 GB HDD

Learning Objectives:

Learn to Perform the data classification algorithm using any Classification algorithm

Outcomes:

After completion of this assignment students are able to understand Perform the data classification algorithm using any Classification algorithm

Theory:

What is Classification?

We use the training dataset to get better boundary conditions which could be used to determine each target class. Once the boundary conditions are determined, the next task is to predict the target class. The whole process is known as classification.

Target class examples:

- Analysis of the customer data to predict whether he will buy computer accessories (Target class: Yes or No)
- Classifying fruits from features like color, taste, size, weight (Target classes: Apple, Orange, Cherry, Banana)
- Gender classification from hair length (Target classes: Male or Female)

Let's understand the concept of classification algorithms with gender classification using hair length (by no means am I trying to stereotype by gender, this is only an example). To classify gender (target class) using hair length as feature parameter we could train a model using any classification algorithms to come up with some set of boundary conditions which can be used to differentiate the male and female genders using hair length as the training feature. In gender classification case the boundary condition could the proper hair length value. Suppose the differentiated boundary hair length value is 15.0 cm then we can say that if hair length is less than 15.0 cm then gender could be male or else female.

Classification Algorithms vs Clustering Algorithms

In clustering, the idea is not to predict the target class as in classification, it's more ever trying to group the similar kind of things by considering the most satisfied condition, all the items in the same group should be similar and no two different group items should not be similar.

Group items Examples:

- While grouping similar language type documents (Same language documents are one group.)
- While categorizing the news articles (Same news category(Sport) articles are one group)

Let's understand the concept with clustering genders based on hair length example. To determine gender, different similarity measure could be used to categorize male and female genders. This could be done by finding the similarity between two hair lengths and keep them in the same group if the similarity is less (Difference of hair length is less). The same process could continue until all the hair length properly grouped into two categories.

Basic Terminology in Classification Algorithms

- Classifier: An algorithm that maps the input data to a specific category.
- Classification model: A classification model tries to draw some conclusion from the input values given for training. It will predict the class labels/categories for the new data.
- Feature: A feature is an individual measurable property of a phenomenon being observed.
- Binary Classification: Classification task with two possible outcomes. Eg: Gender classification (Male / Female)
- Multi-class classification: Classification with more than two classes. In multi-class classification, each sample is assigned to one and only one target label. Eg: An animal can be a cat or dog but not both at the same time.
- Multi-label classification: Classification task where each sample is mapped to a set of target labels (more than one class). Eg: A news article can be about sports, a person, and location at the same time.

Applications of Classification Algorithms

- Email spam classification
- Bank customers loan pay willingness prediction.
- Cancer tumor cells identification.
- Sentiment analysis
- Drugs classification
- Facial key points detection
- Pedestrians detection in an automotive car driving.

Types of Classification Algorithms

Classification Algorithms could be broadly classified as the following:

- Linear Classifiers
  - Logistic regression
  - Naive Bayes classifier
  - Fisher's linear discriminant
- Support vector machines
  - Least squares support vector machines
- Quadratic classifiers
- Kernel estimation
  - k-nearest neighbor
- Decision trees
  - Random forests
- Neural networks
- Learning vector quantization

Consider the annual rainfall details at a place starting from January 2012. We create an R time series object for a period of 12 months and plot it.

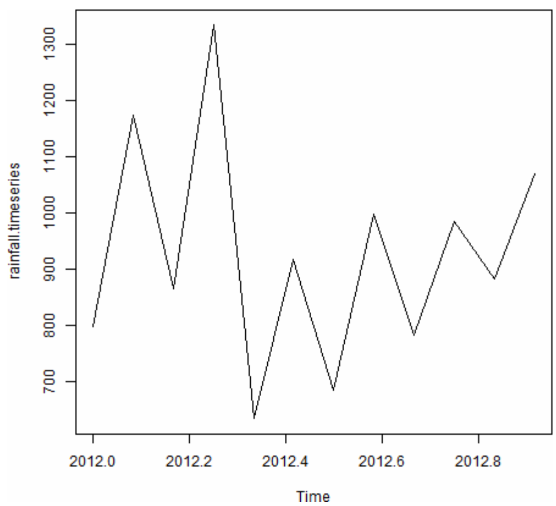
```
# Get the data points in form of a R vector.rainfall <-
c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)

# Convert it to a time series object.
rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)

# Print the timeseries
data.
print(rainfall.timeseries)

# Give the chart file a name.png(file =
"rainfall.png")

# Plot a graph of the time series.
plot(rainfall.timeseries)
```

# Save the file.
dev.off()

Output:

When we execute the above code, it produces the following result and chart −

```
Jan        Feb        Mar        Apr        May        Jun        Jul        Aug
 Sep 2012 799.0     1174.8 865.1 1334.6 635.4 918.5 685.5 998.6 784.2
              Oct          Nov         Dec
 2012      985.0        882.8 1071.0
```



Conclusion:- Thus, this way Performed data classification algorithm using any Classification algorithm

Assignment No. 6

Title:

Perform Data Clustering Using Any Clustering Algorithm

Objectives:

- To understand the concept of clustering in machine learning
- To implement a clustering algorithm on a dataset
- To analyze grouping of similar data points

Problem Statement:

Perform data clustering using any clustering algorithm and group similar data points based on their features.

Software Requirements:

- Python (Anaconda / Jupyter Notebook) or Google Colab
- Libraries: NumPy, Matplotlib, Scikit-learn

Hardware Requirements:

- Processor: Minimum 1 GHz
- Hard Drive: Minimum 32 GB
- Memory (RAM): Minimum 2 GB

Theory

Introduction

Clustering is an unsupervised machine learning technique used to group similar data points into clusters based on their characteristics. Unlike classification, clustering does not use labeled data. It helps in identifying hidden patterns or structures within data.

Clustering Concept

In clustering:

- Data points in the same cluster are highly similar
- Data points in different clusters are dissimilar

Clustering is widely used in data analysis, pattern recognition, and business intelligence.

Clustering vs Classification

- **Clustering:** Unsupervised learning, no predefined labels
- **Classification:** Supervised learning, uses labeled data

K-Means Clustering Algorithm

K-Means is a popular clustering algorithm that partitions the dataset into **K clusters**. Each cluster is represented by its centroid (mean value of points in that cluster).

Steps of K-Means Algorithm

1. Choose the number of clusters (K)
2. Initialize centroids randomly
3. Assign each data point to the nearest centroid
4. Update centroids by calculating mean
5. Repeat until convergence

Applications of Clustering

- Customer segmentation

- Market analysis
- Image segmentation
- Recommendation systems
- Anomaly detection

Implementation / Procedure:

- Import required libraries
- Load or create dataset
- Apply K-Means clustering algorithm
- Fit the model to the data
- Predict cluster labels
- Visualize clusters using graphs

Python Implementation

```
# Import libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Dataset
X = np.array([[1,2], [1,4], [1,0],
              [10,2], [10,4], [10,0]])

# Apply K-Means
kmeans = KMeans(n_clusters=2)
kmeans.fit(X)

# Results
labels = kmeans.labels_
centroids = kmeans.cluster_centers_

# Visualization
plt.scatter(X[:,0], X[:,1], c=labels)
plt.scatter(centroids[:,0], centroids[:,1], marker='X')
plt.title("K-Means Clustering")
plt.show()
```

Configuration Steps:

- Install required libraries (NumPy, Matplotlib, Scikit-learn)
- Set up Python environment (Jupyter Notebook / Colab)

- Load dataset properly
- Configure number of clusters (K value)
- Execute code and verify output

Conclusion:

In this assignment, we successfully implemented a clustering algorithm using K-Means. The algorithm grouped similar data points into clusters based on their features. Clustering is useful for discovering hidden patterns and plays an important role in data analysis and machine learning applications.